

Active Directory e Linux

- Installiamo Ubuntu 20.04
- Qualche piccola configurazione
- Installiamo il pacchetto sssd (System Security Services Daemon) e altro
- Verifichiamo che tutto funzioni!!!

Active Directory e Linux

Come abbiamo già fatto per l'installazione di Windows Server 2019, creiamo una nuova VM.

Linux è meno esigente in termini di risorse, quindi possiamo assegnare a questa macchina un solo gigabyte di RAM.

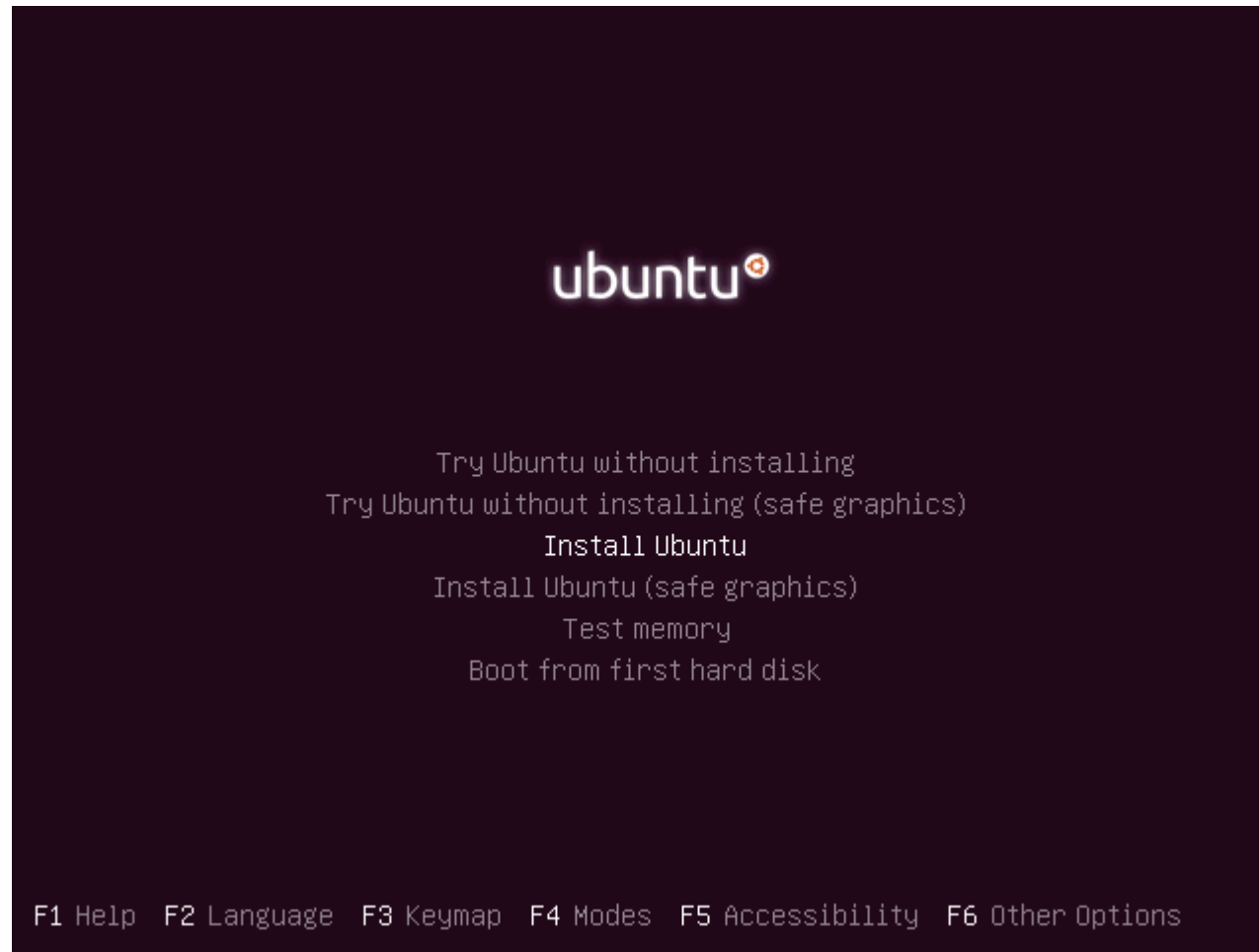
Utilizzeremo la distribuzione Linux Ubuntu 20.04 LTS

Active Directory e Linux

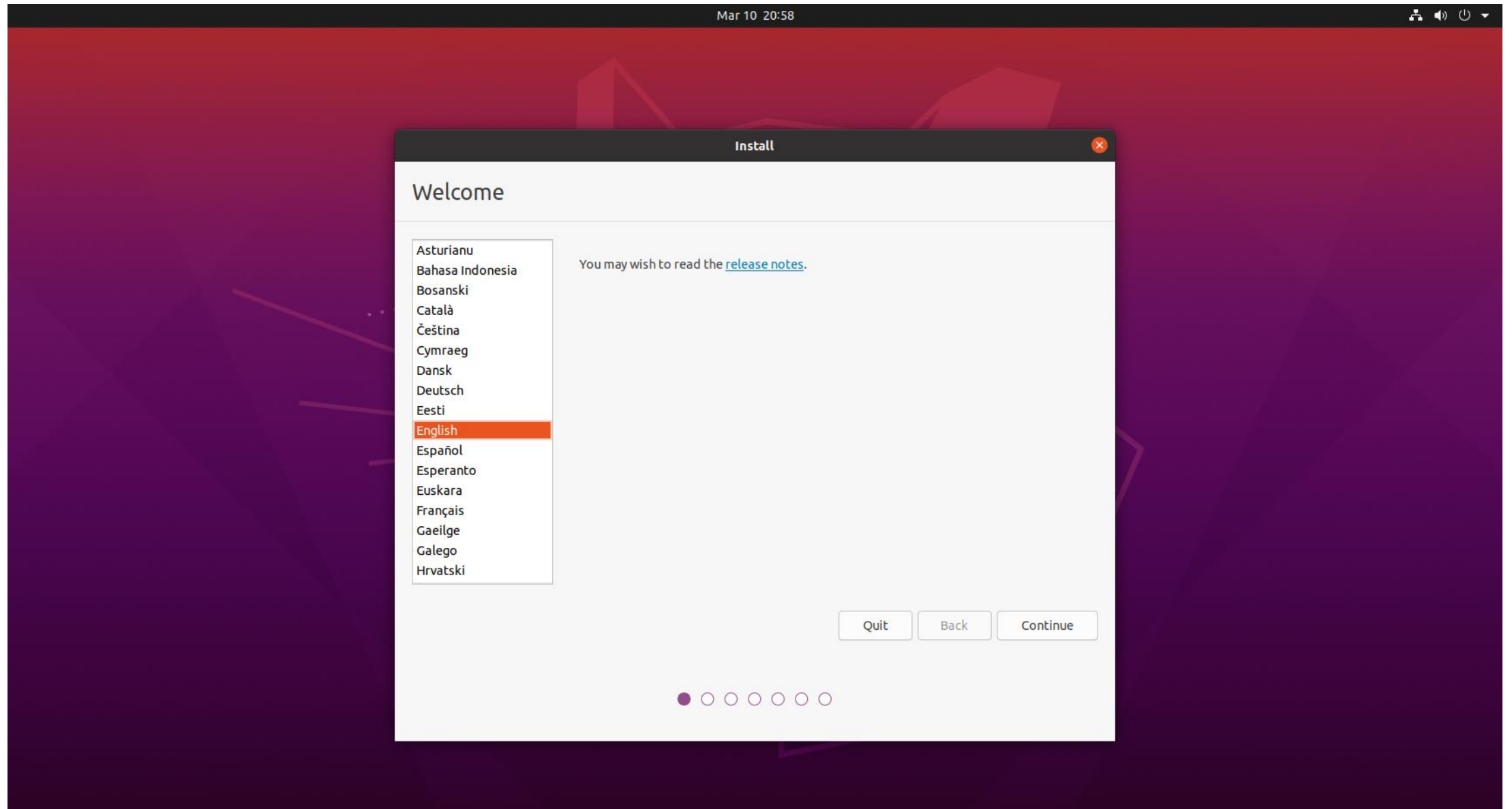
Language			
Amharic	Français	Makedonски	Tamil
Arabic	Gaeilge	Malayalam	தமிழ்
Asturianu	Galego	Marathi	Тоик
Беларуская	Gujarati	Burmese	Thai
Български	עברית	Nepali	Tagalog
Bengali	Hindi	Nederlands	Türkçe
Tibetan	Hrvatski	Norsk bokmål	Uyghur
Bosanski	Magyar	Norsk nynorsk	Українська
Català	Bahasa Indonesia	Punjabi (Gurmukhi)	Tiếng Việt
Čeština	Íslenska	Polski	中文(简体)
Dansk	Italiano	Português do Brasil	中文(繁體)
Deutsch	日本語	Português	
Dzongkha	ქართული	Română	
Ελληνικά	Қазақ	Русский	
English	Khmer	Sámegiellii	
Esperanto	ಕನ್ನಡ	සිංහල	
Español	한국어	Slovenčina	
Eesti	Kurdî	Slovensčina	
Euskara	Lao	Shqip	
فارسی	Lietuviškai	Српски	
Suomi	Latviski	Svenska	

F1 Help F2 Language F3 Keymap F4 Modes F5 Accessibility F6 Other Options

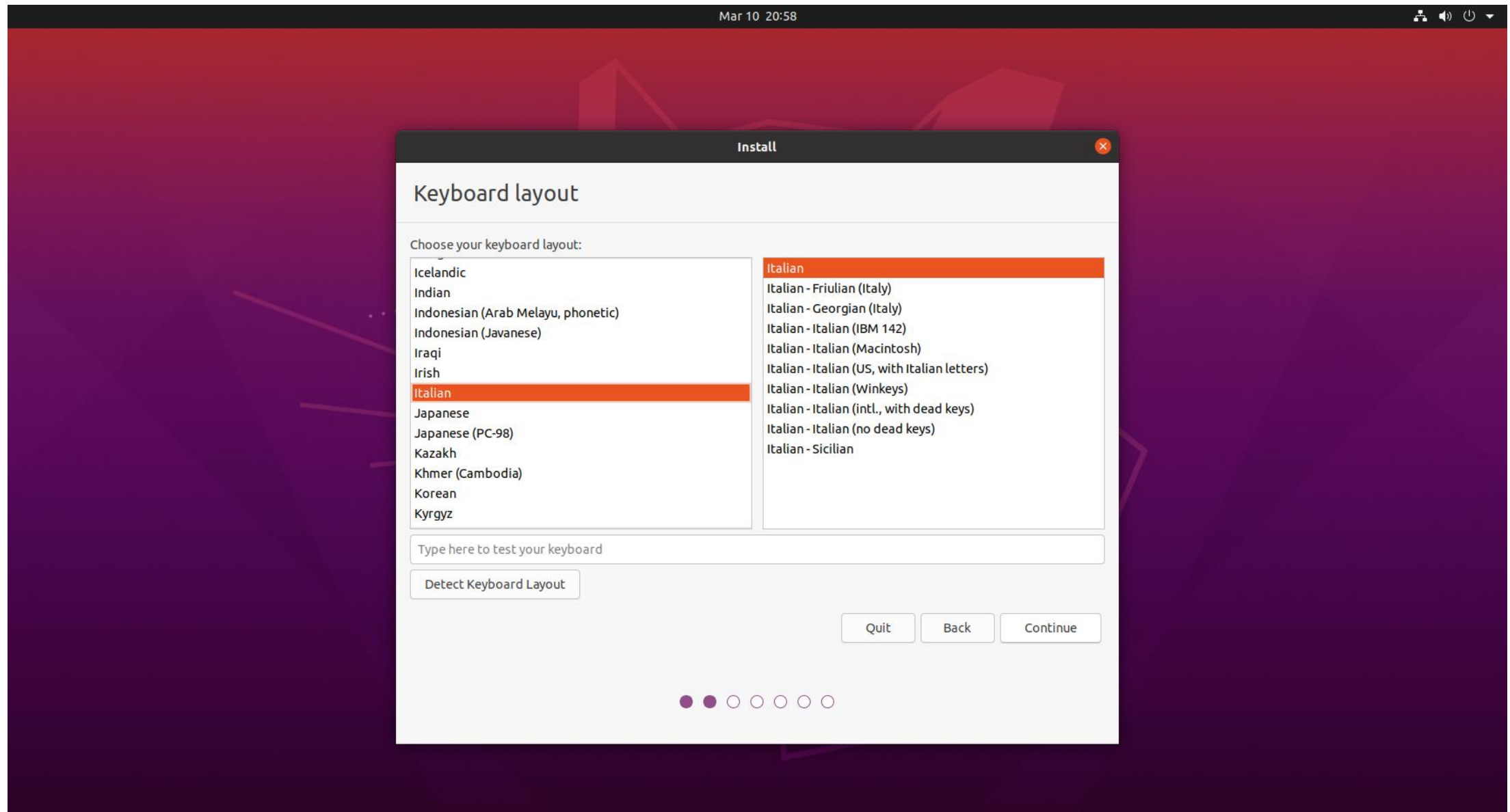
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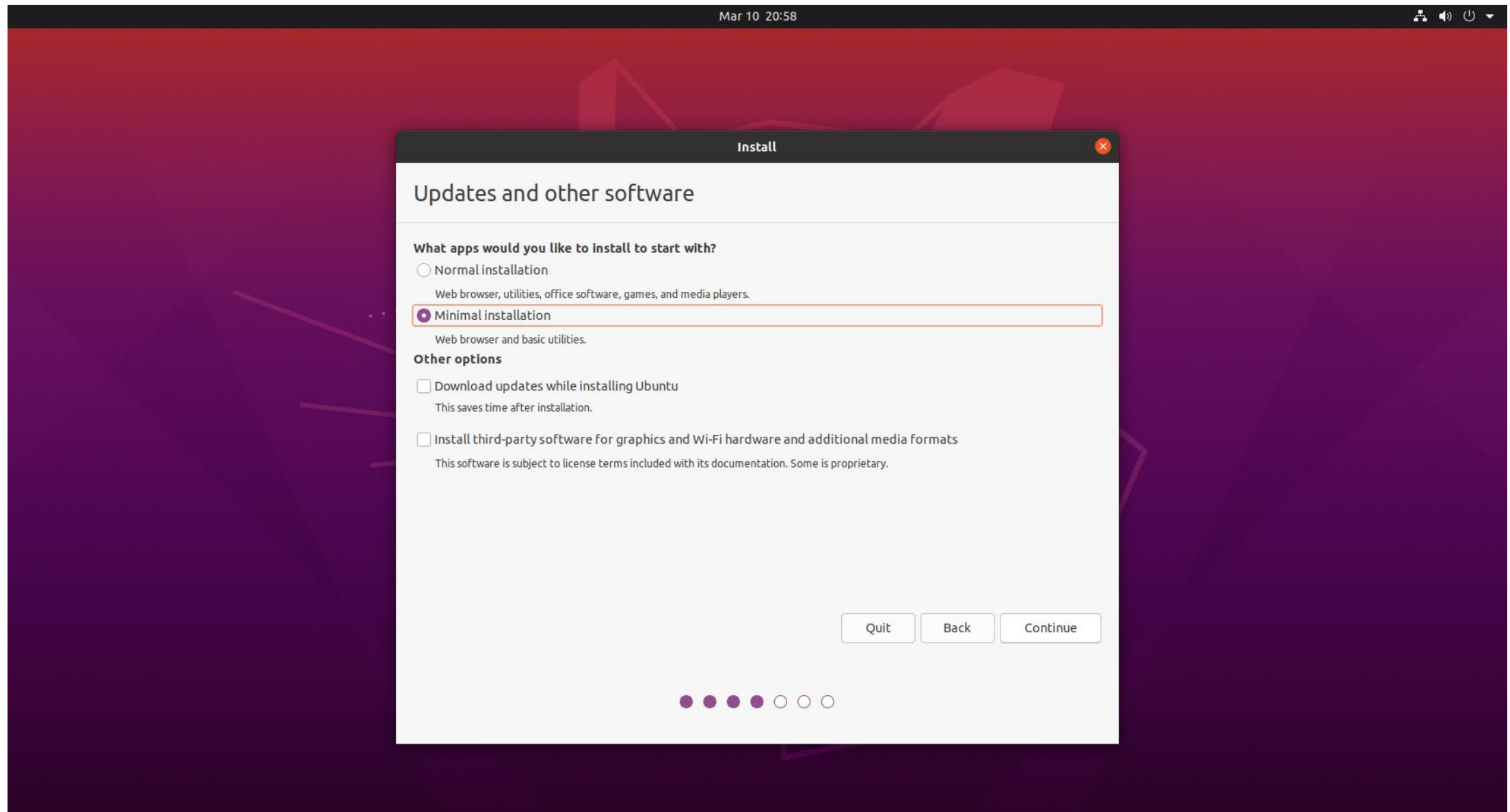
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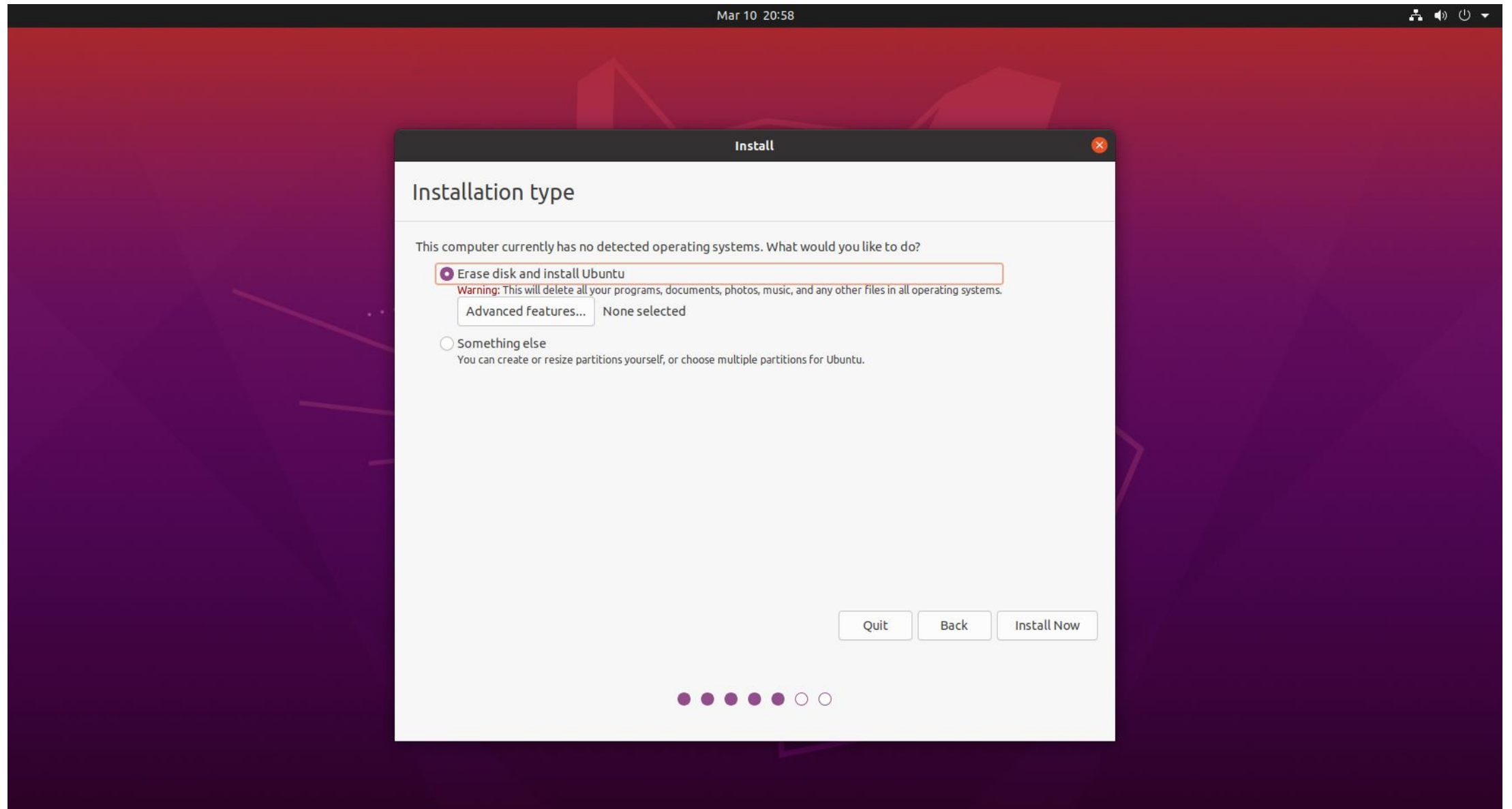
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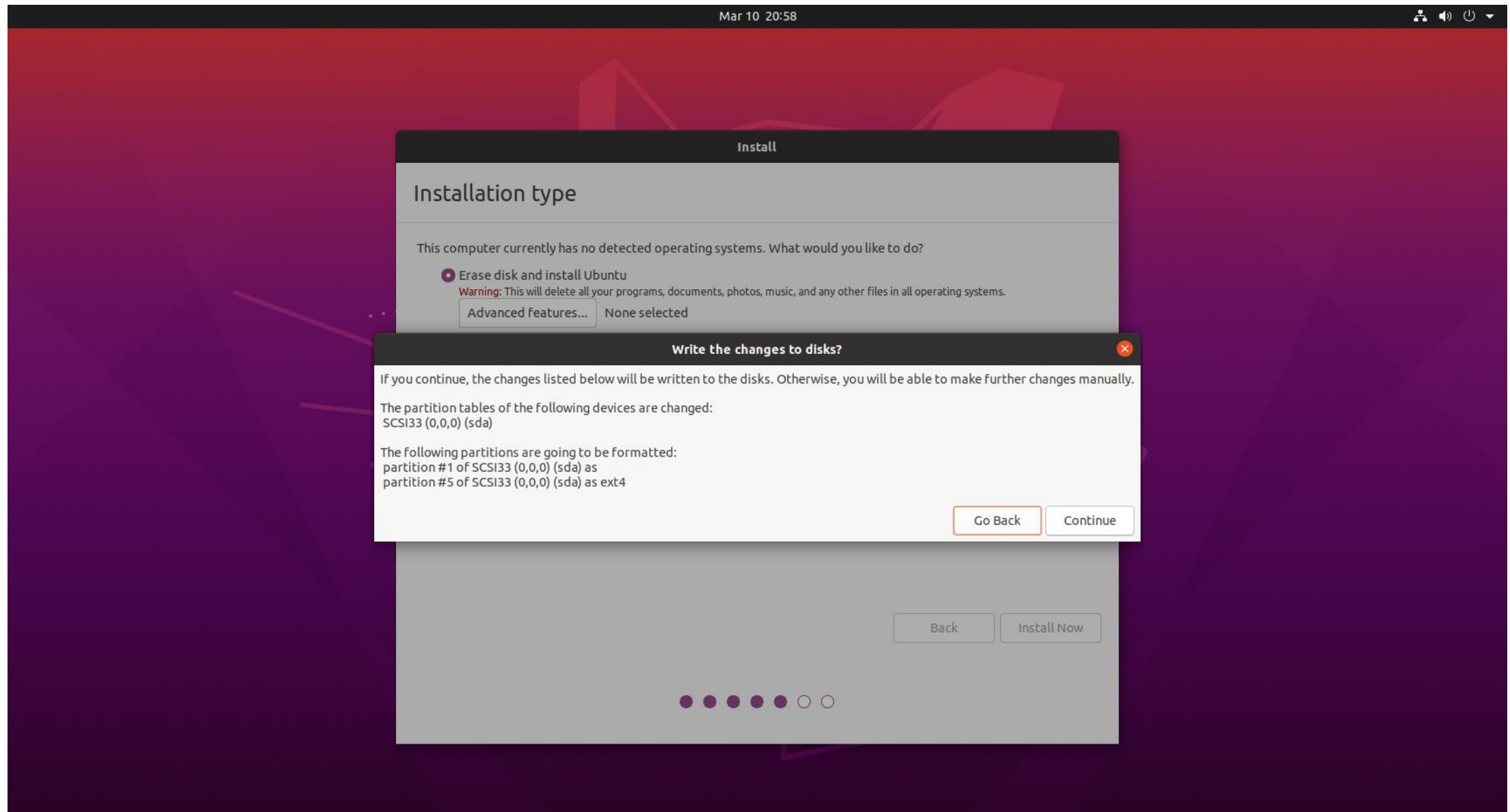
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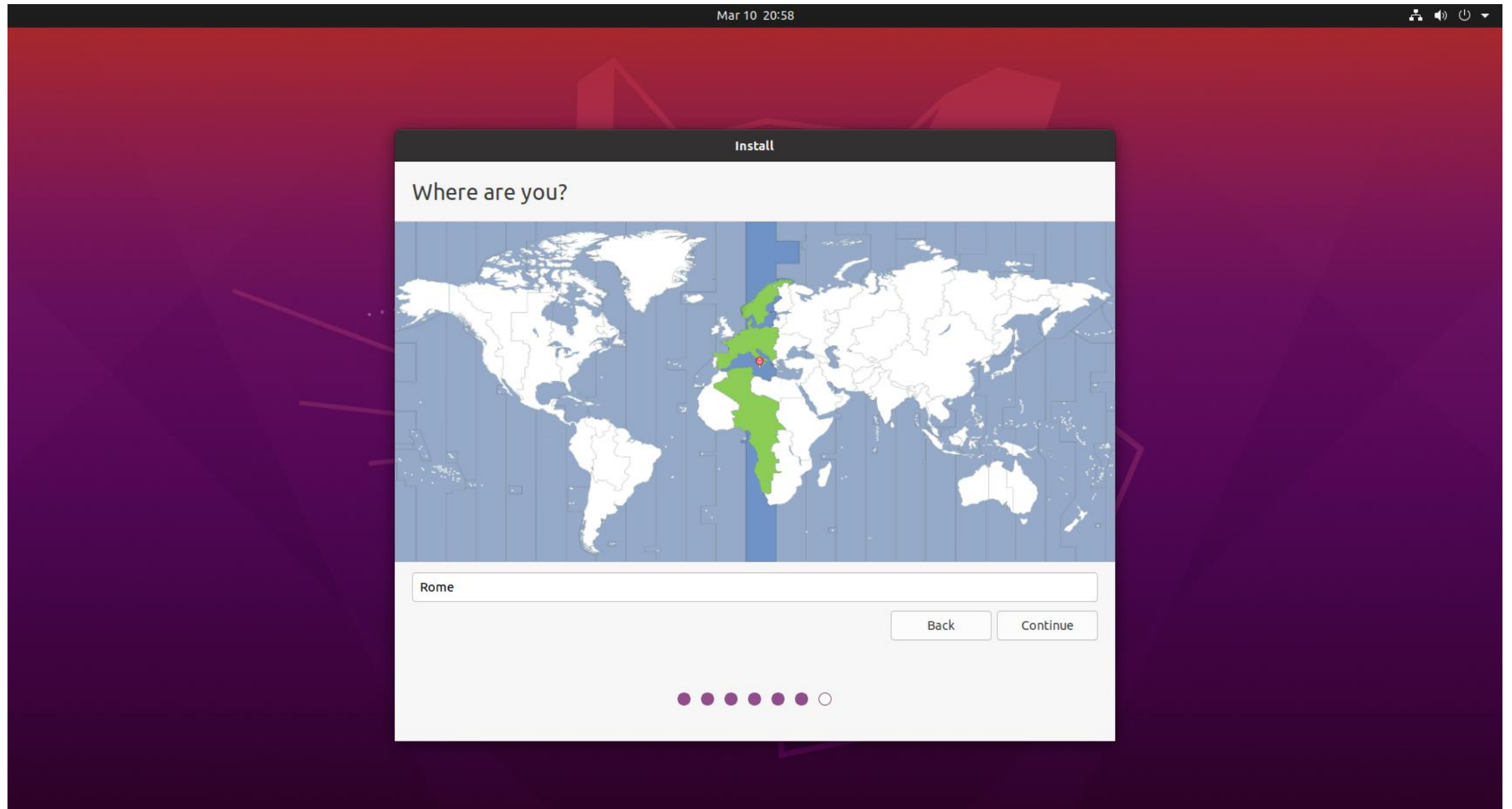
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Mar 10 21:59

Install

Who are you?

Your name: administrator ✓

Your computer's name: LinuxClient ✓
The name it uses when it talks to other computers.

Pick a username: administrator ✓

Choose a password: SysInt00 Fair password

Confirm your password: SysInt00 ✓

☐ Log in automatically

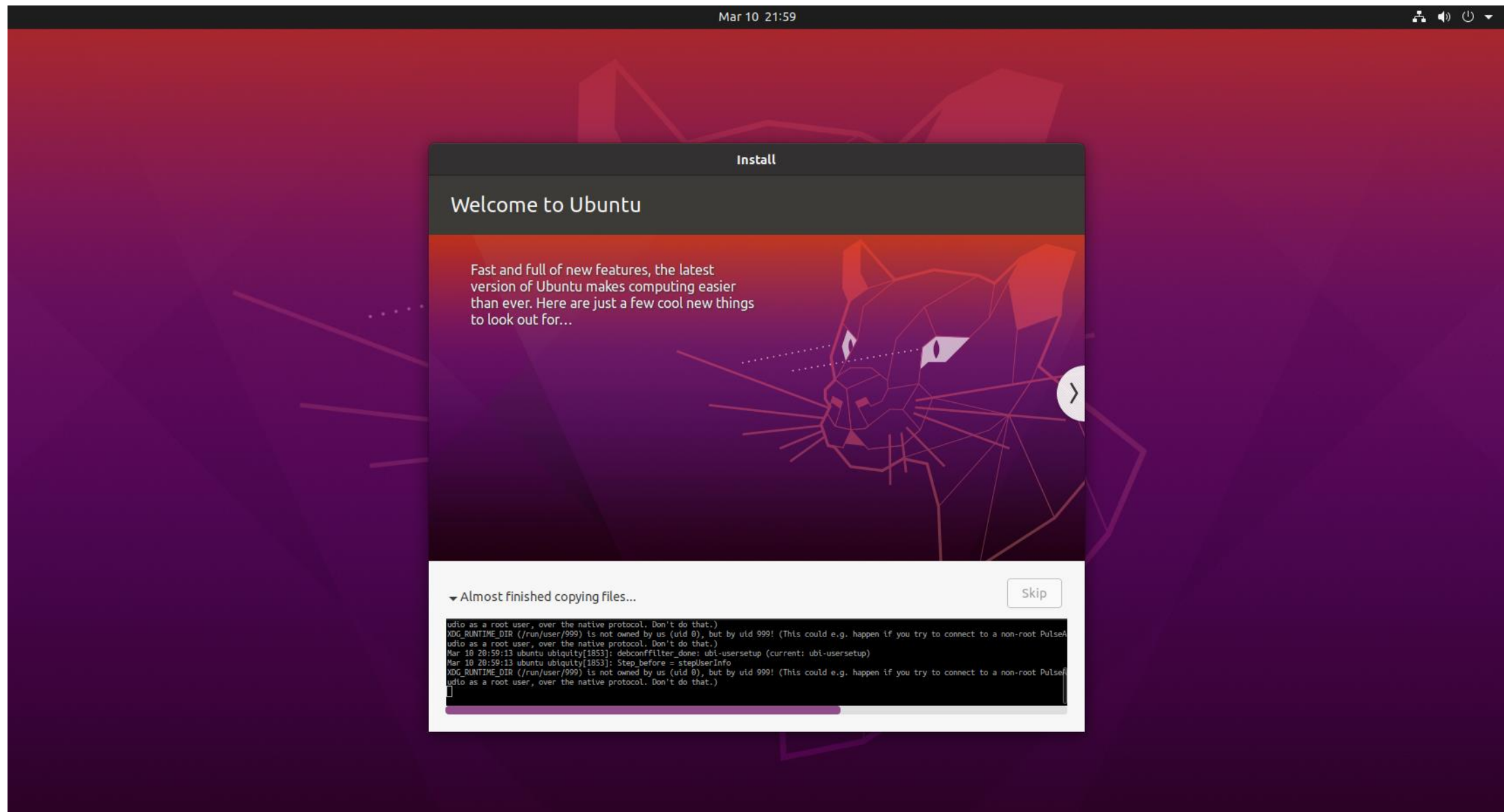
☒ Require my password to log in

☐ Use Active Directory
You'll enter domain and other details in the next step.

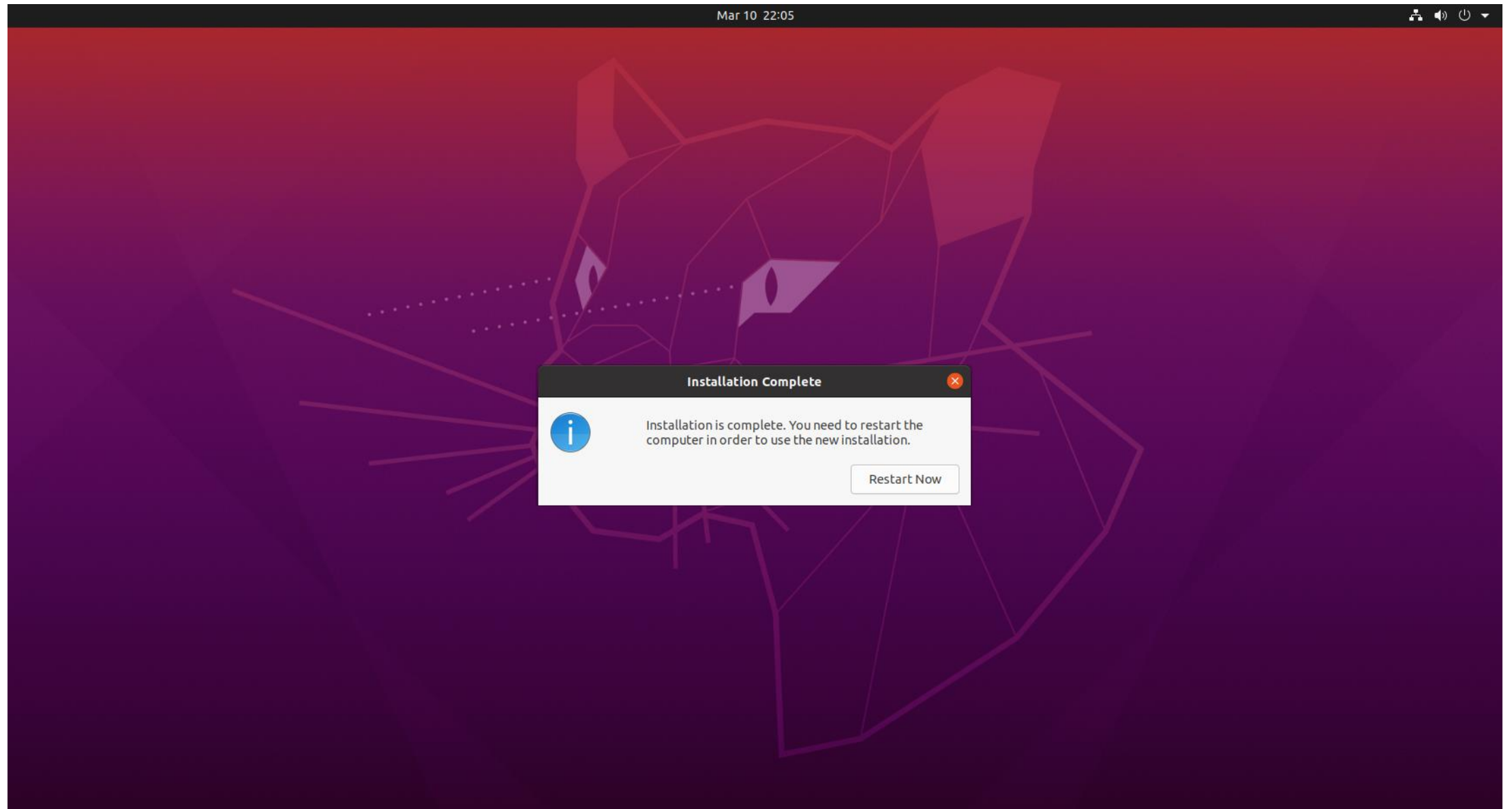
Back Continue

● ● ● ● ● ● ● ●

Active Directory e Linux



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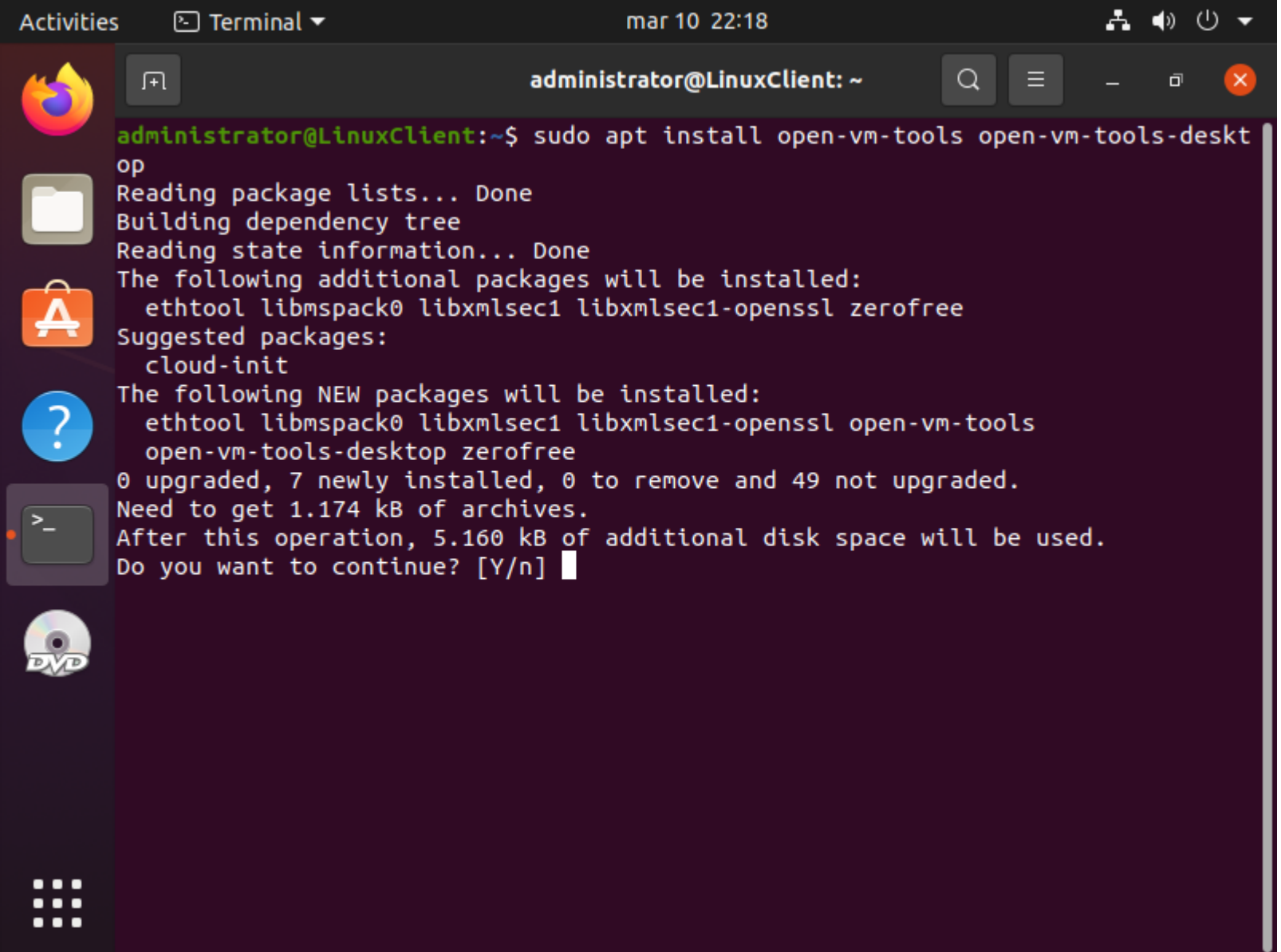
Active Directory e Linux

Ora installiamo i VMware Tools

Su Linux abbiamo due possibilità:

1. Installare i tools che ci mette a disposizione VMware
2. Installare dei tools open che ci vengono forniti dalla distribuzione

Questa seconda opzione è preferibile in quanto mantiene i tools aggiornati insieme al resto dei pacchetti.



```
Activities Terminal mar 10 22:18
administrator@LinuxClient: ~
administrator@LinuxClient:~$ sudo apt install open-vm-tools open-vm-tools-desktop
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  ethtool libmspack0 libxmlsec1 libxmlsec1-openssl zerofree
Suggested packages:
  cloud-init
The following NEW packages will be installed:
  ethtool libmspack0 libxmlsec1 libxmlsec1-openssl open-vm-tools
  open-vm-tools-desktop zerofree
0 upgraded, 7 newly installed, 0 to remove and 49 not upgraded.
Need to get 1.174 kB of archives.
After this operation, 5.160 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

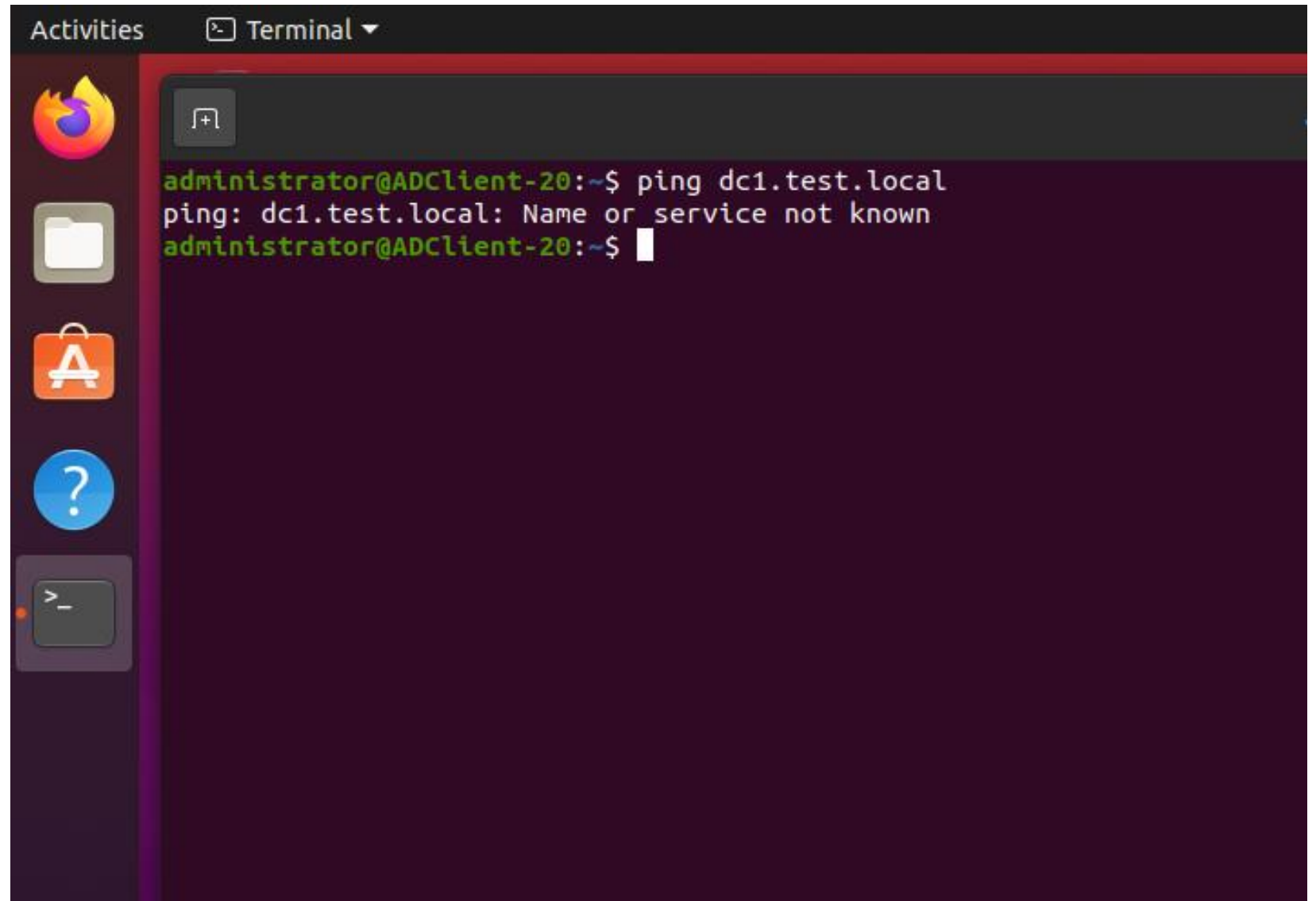
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```
Activities Terminal mar 10 22:18 administrator@LinuxClient: ~
Get:5 http://it.archive.ubuntu.com/ubuntu focal/main amd64 ethtool amd64 1:5.4-1 [134 kB]
Get:6 http://it.archive.ubuntu.com/ubuntu focal-updates/universe amd64 open-vm-tools-desktop amd64 2:11.3.0-2ubuntu0~ubuntu20.04.2 [144 kB]
Get:7 http://it.archive.ubuntu.com/ubuntu focal/main amd64 zerofree amd64 1.1.1-1 [8.520 B]
Fetched 1.174 kB in 0s (7.172 kB/s)
Selecting previously unselected package libmspack0:amd64.
(Reading database ... 155465 files and directories currently installed.)
Preparing to unpack .../0-libmspack0_0.10.1-2_amd64.deb ...
Unpacking libmspack0:amd64 (0.10.1-2) ...
Selecting previously unselected package libxmlsec1:amd64.
Preparing to unpack .../1-libxmlsec1_1.2.28-2_amd64.deb ...
Unpacking libxmlsec1:amd64 (1.2.28-2) ...
Selecting previously unselected package libxmlsec1-openssl:amd64.
Preparing to unpack .../2-libxmlsec1-openssl_1.2.28-2_amd64.deb ...
Unpacking libxmlsec1-openssl:amd64 (1.2.28-2) ...
Selecting previously unselected package open-vm-tools.
Preparing to unpack .../3-open-vm-tools_2%3a11.3.0-2ubuntu0~ubuntu20.04.2_amd64.deb ...
Unpacking open-vm-tools (2:11.3.0-2ubuntu0~ubuntu20.04.2) ...
Selecting previously unselected package ethtool.
Preparing to unpack .../4-ethtool_1%3a5.4-1_amd64.deb ...
Unpacking ethtool (1:5.4-1) ...
Selecting previously unselected package open-vm-tools-desktop.
Preparing to unpack .../5-open-vm-tools-desktop_2%3a11.3.0-2ubuntu0~ubuntu20.04.2_amd64.deb ...
Unpacking open-vm-tools-desktop (2:11.3.0-2ubuntu0~ubuntu20.04.2) ...
Selecting previously unselected package zerofree.
Preparing to unpack .../6-zerofree_1.1.1-1_amd64.deb ...
Unpacking zerofree (1.1.1-1) ...
Setting up zerofree (1.1.1-1) ...
Setting up libmspack0:amd64 (0.10.1-2) ...
Setting up libxmlsec1:amd64 (1.2.28-2) ...
Setting up ethtool (1:5.4-1) ...
Setting up libxmlsec1-openssl:amd64 (1.2.28-2) ...
Setting up open-vm-tools (2:11.3.0-2ubuntu0~ubuntu20.04.2) ...
Created symlink /etc/systemd/system/vmtoolsd.service → /lib/systemd/system/open-vm-tools.service.
Created symlink /etc/systemd/system/multi-user.target.wants/open-vm-tools.service → /lib/systemd/system/open-vm-tools.service.
Created symlink /etc/systemd/system/open-vm-tools.service.requires/vgauth.service → /lib/systemd/system/vgauth.service.
Setting up open-vm-tools-desktop (2:11.3.0-2ubuntu0~ubuntu20.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/run-vmblock\x2dfuse.mount → /lib/systemd/system/run-vmblock\x2dfuse.mount.
Processing triggers for libc-bin (2.31-0ubuntu9.2) ...
Processing triggers for systemd (245.4-4ubuntu3.15) ...
Processing triggers for man-db (2.9.1-1) ...
Progress: [ 97%] [#####.....]
```


Active Directory e Linux

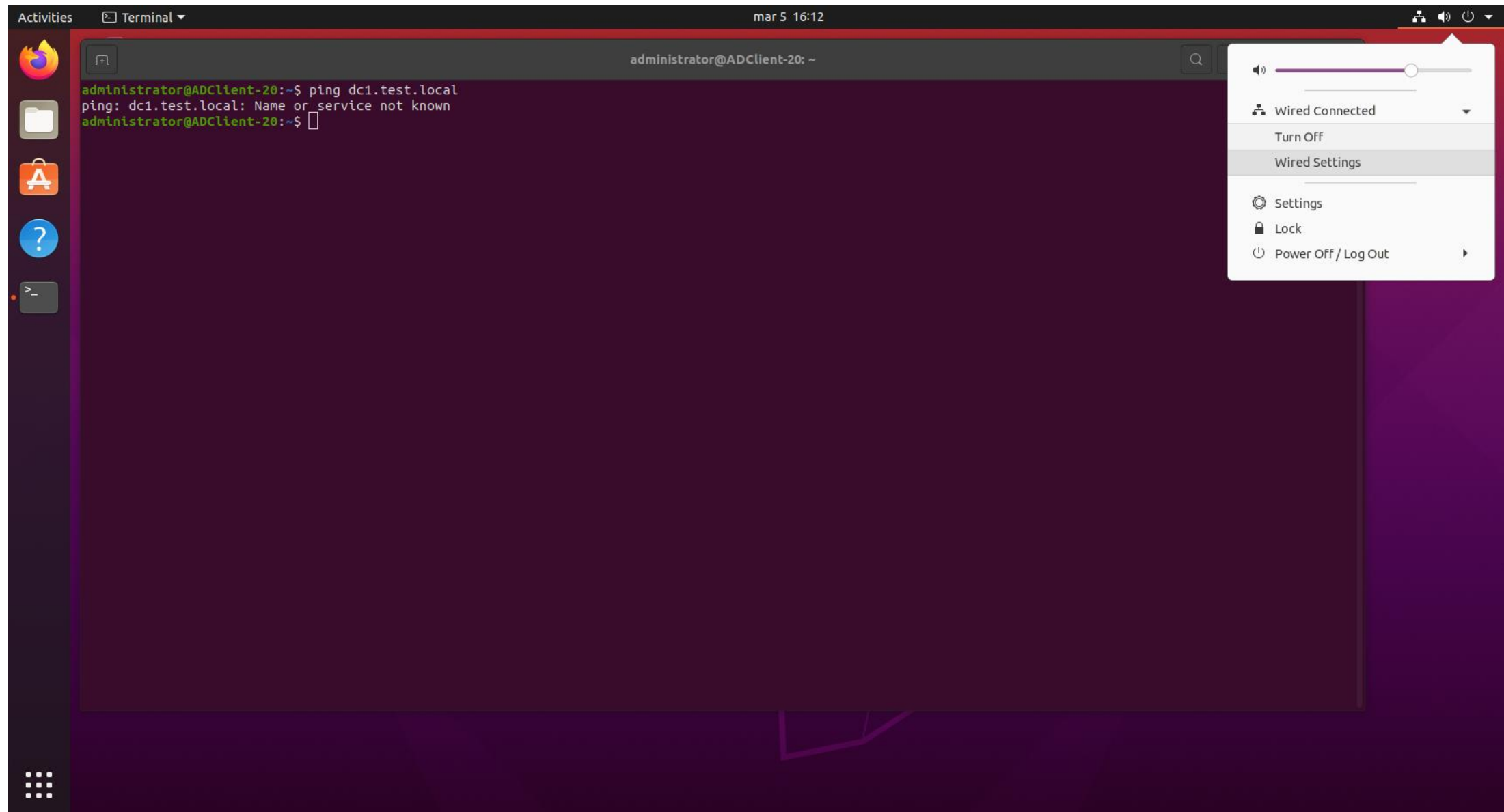
Pinghiamo il domain controller

Perché non funziona?

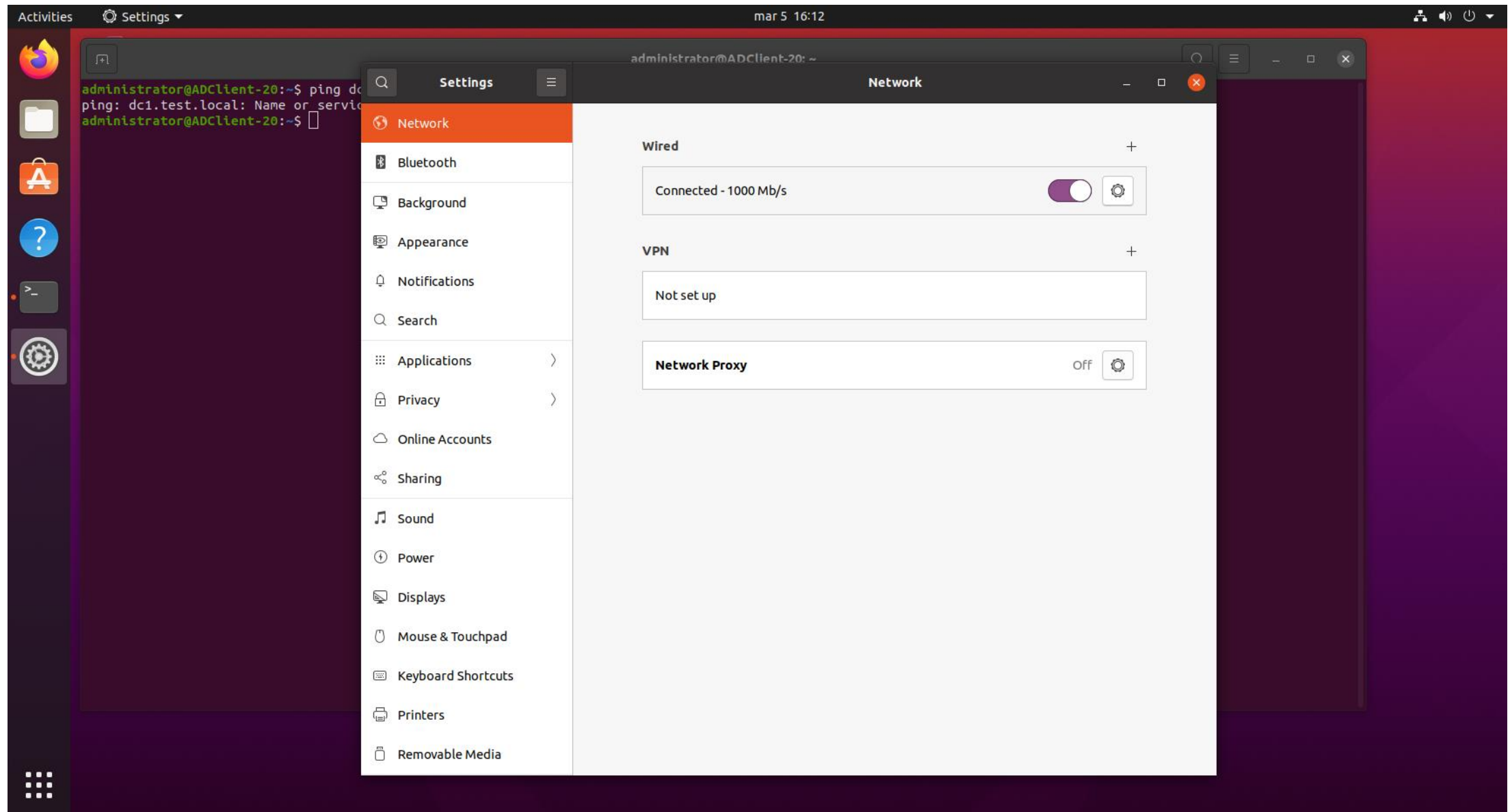
A screenshot of a Linux desktop environment. The top bar shows 'Activities' and 'Terminal'. On the left is a dock with icons for Firefox, Files, App Store, a question mark, and a terminal icon. The terminal window is open, showing a command prompt 'administrator@ADClient-20:~\$' followed by 'ping dc1.test.local'. The output is 'ping: dc1.test.local: Name or service not known'. The prompt is followed by another 'administrator@ADClient-20:~\$' and a cursor.

```
administrator@ADClient-20:~$ ping dc1.test.local
ping: dc1.test.local: Name or service not known
administrator@ADClient-20:~$
```

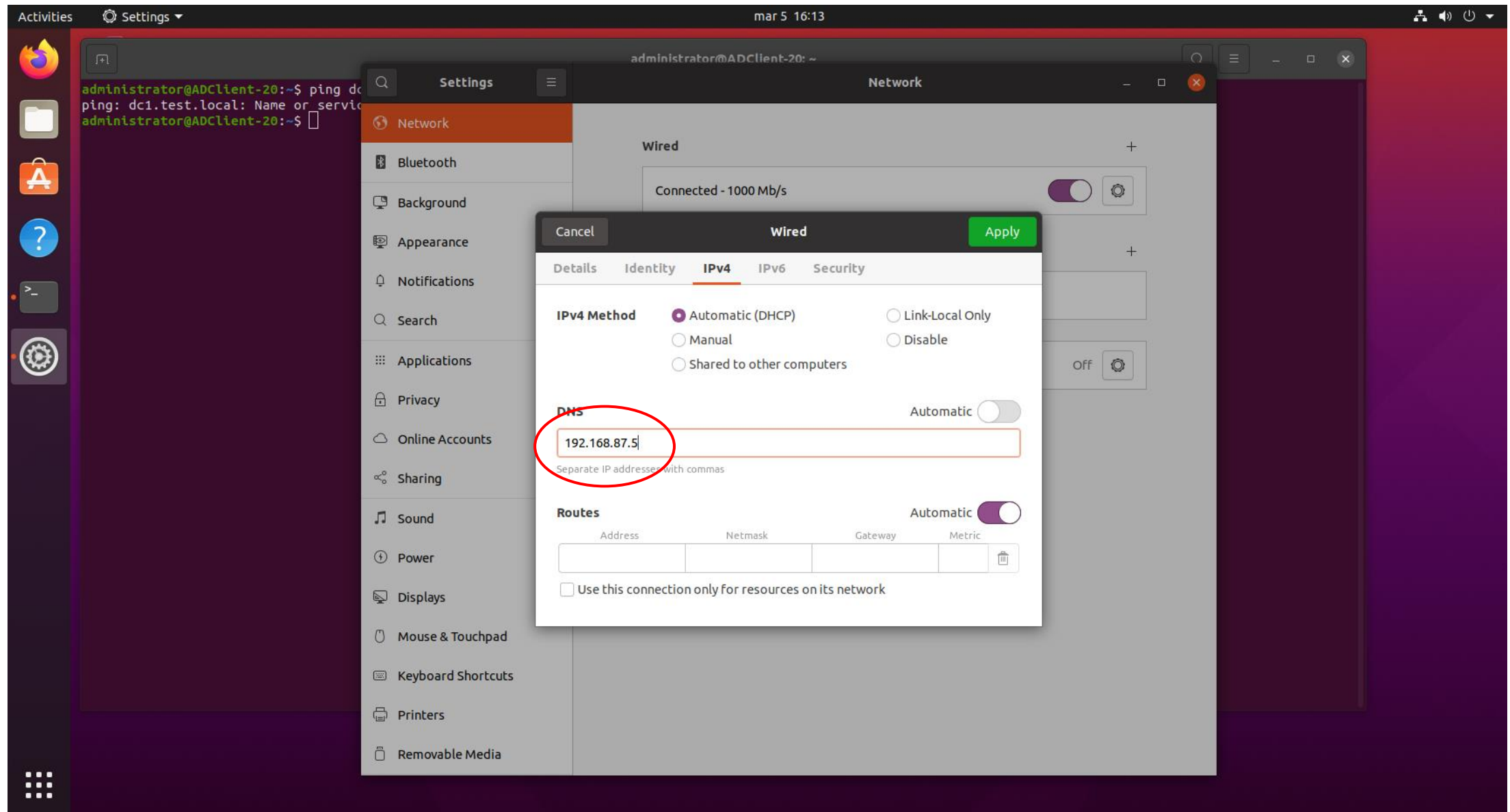

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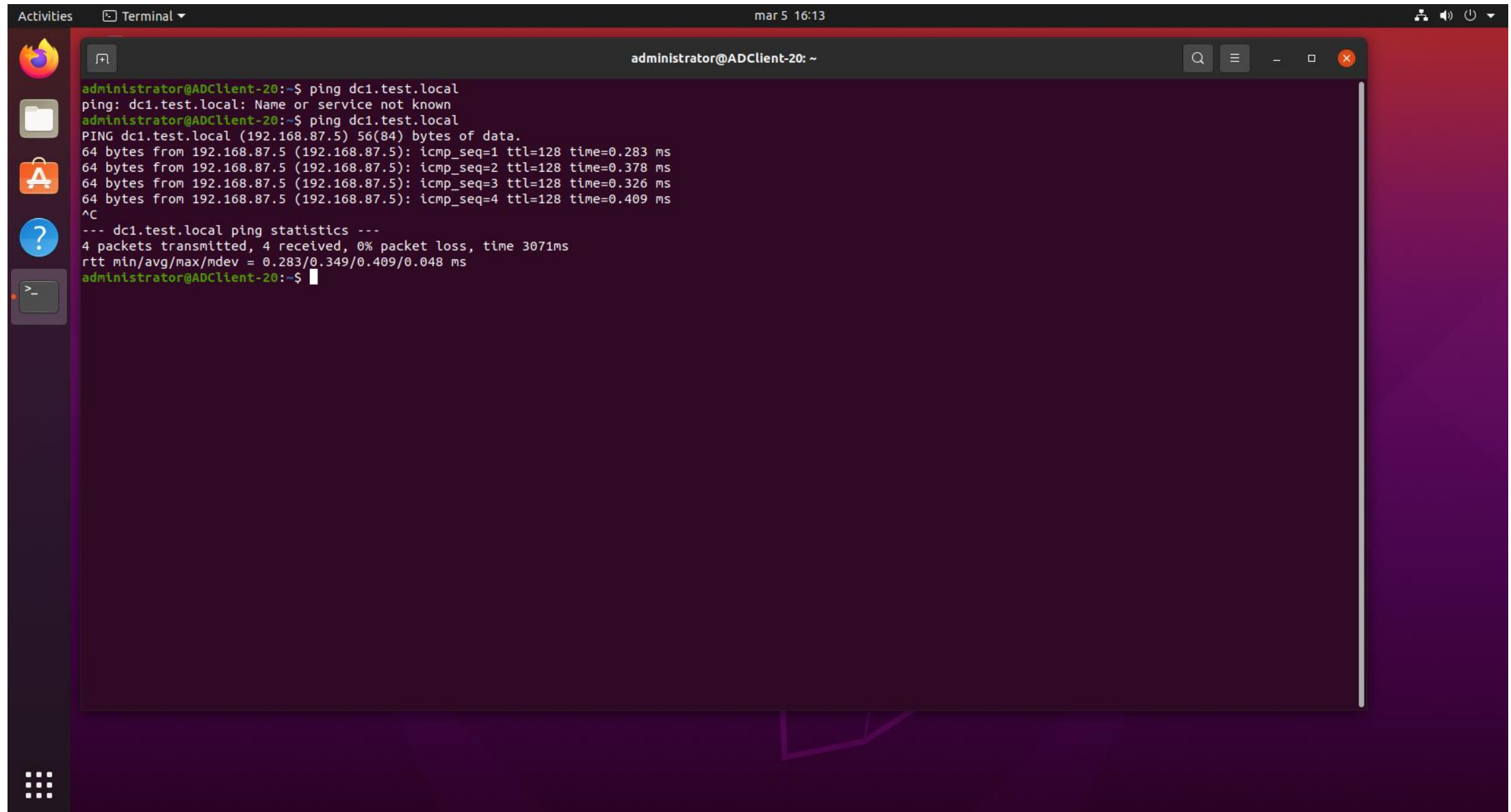
Active Directory e Linux



Active Directory e Linux



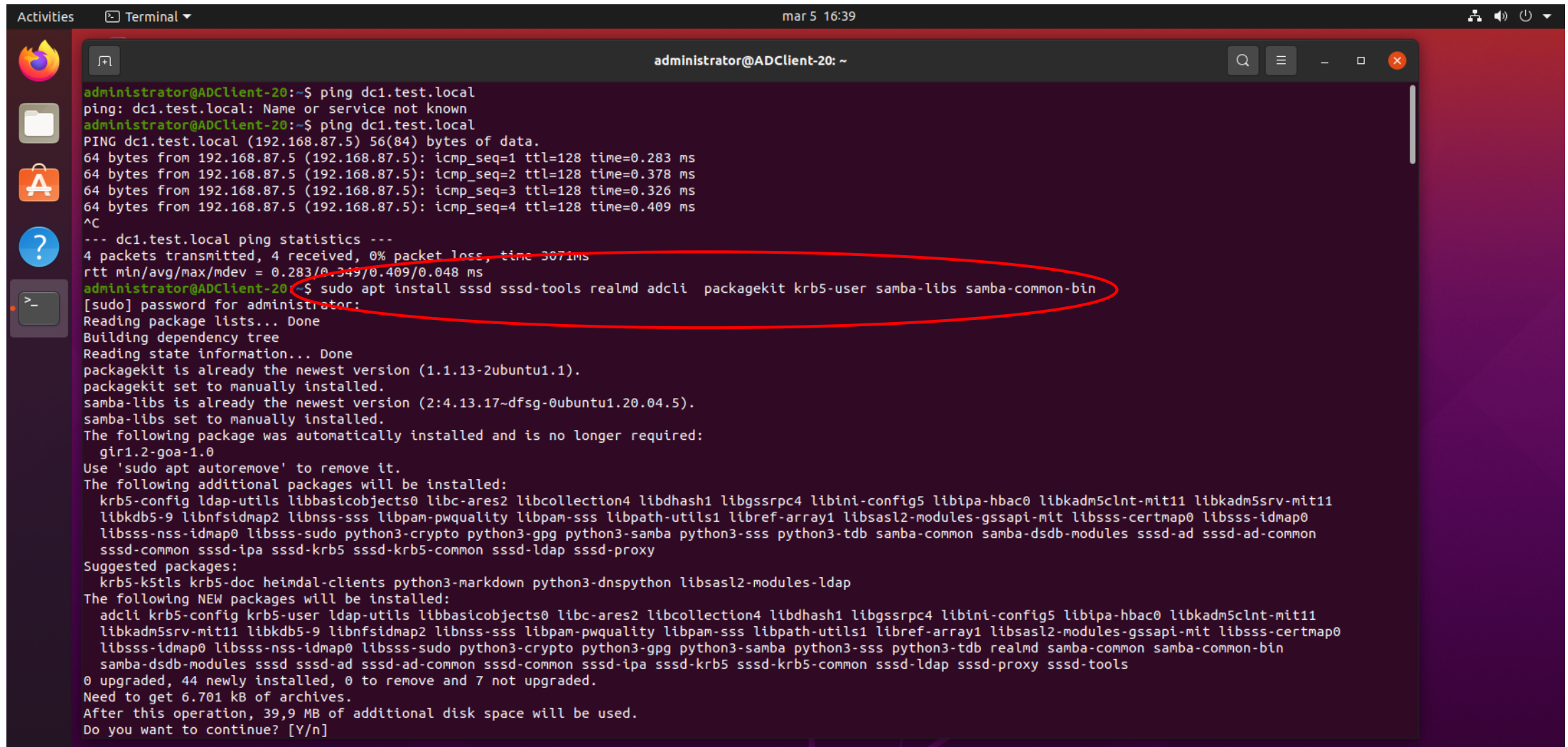
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The screenshot shows a Linux desktop environment with a terminal window open. The terminal title bar reads "administrator@ADClient-20: ~". The terminal output shows a ping command being executed against "dc1.test.local". The first attempt fails with the message "ping: dc1.test.local: Name or service not known". The second attempt is successful, showing four successful ping responses with varying times. The terminal also displays the ping statistics for the successful command.

```
administrator@ADClient-20:~$ ping dc1.test.local
ping: dc1.test.local: Name or service not known
administrator@ADClient-20:~$ ping dc1.test.local
PING dc1.test.local (192.168.87.5) 56(84) bytes of data.
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=1 ttl=128 time=0.283 ms
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=2 ttl=128 time=0.378 ms
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=3 ttl=128 time=0.326 ms
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=4 ttl=128 time=0.409 ms
^C
--- dc1.test.local ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3071ms
rtt min/avg/max/mdev = 0.283/0.349/0.409/0.048 ms
administrator@ADClient-20:~$
```

Active Directory e Linux

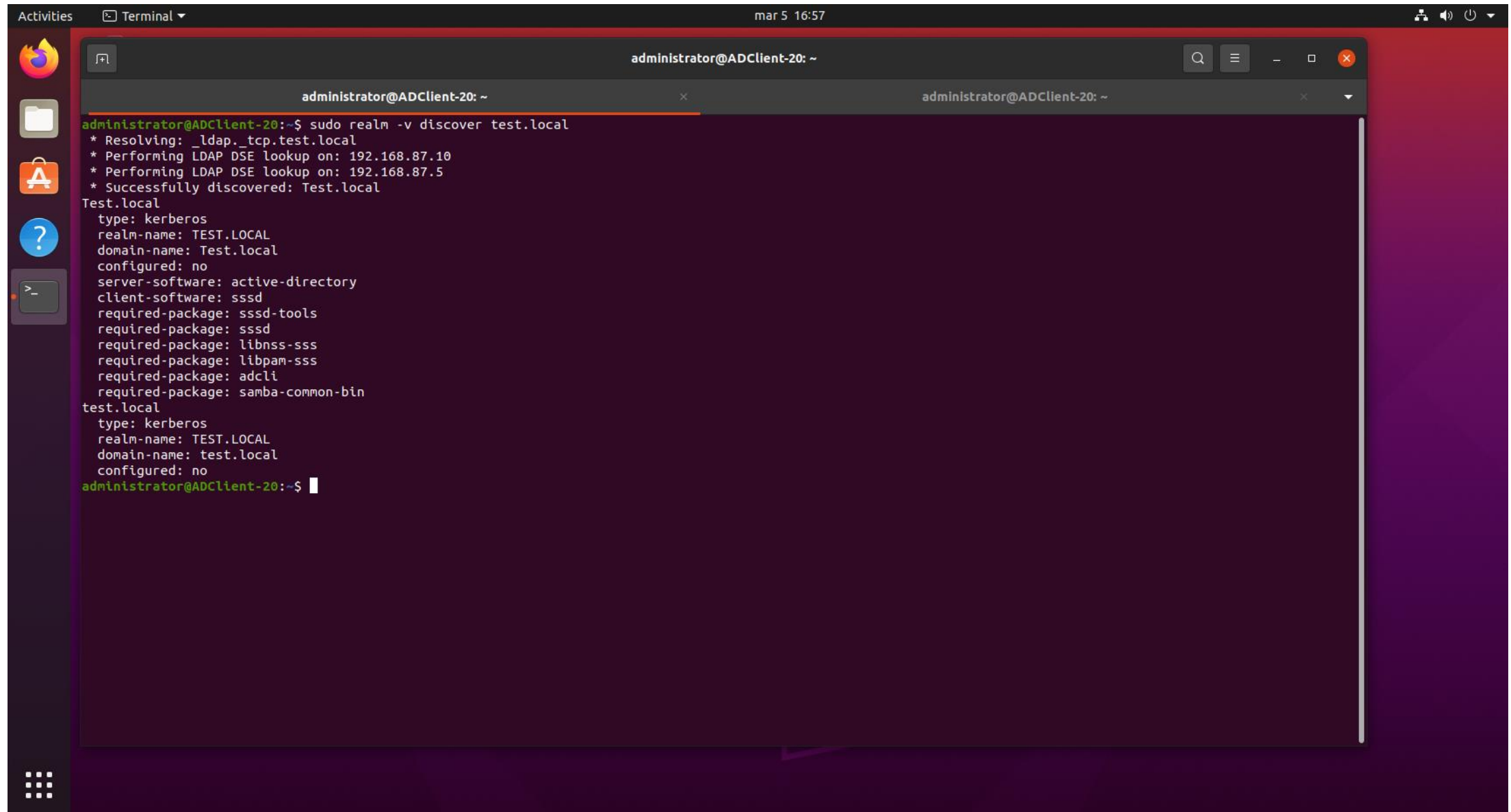


The screenshot shows a terminal window titled 'administrator@ADClient-20: ~'. The terminal output includes a ping command to 'dc1.test.local' which fails with 'Name or service not known'. A second ping command is executed, showing successful results with 0% packet loss and a time of 0.307ms. The command 'sudo apt install sssd sssd-tools realmd adcli packagekit krb5-user samba-libs samba-common-bin' is entered and highlighted with a red oval. The terminal output shows the progress of the installation, including reading package lists, building a dependency tree, and listing additional packages to be installed.

```
administrator@ADClient-20:~$ ping dc1.test.local
ping: dc1.test.local: Name or service not known
administrator@ADClient-20:~$ ping dc1.test.local
PING dc1.test.local (192.168.87.5) 56(84) bytes of data.
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=1 ttl=128 time=0.283 ms
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=2 ttl=128 time=0.378 ms
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=3 ttl=128 time=0.326 ms
64 bytes from 192.168.87.5 (192.168.87.5): icmp_seq=4 ttl=128 time=0.409 ms
^C
--- dc1.test.local ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 307ms
rtt min/avg/max/mdev = 0.283/0.349/0.409/0.048 ms
administrator@ADClient-20:~$ sudo apt install sssd sssd-tools realmd adcli packagekit krb5-user samba-libs samba-common-bin
[sudo] password for administrator:
Reading package lists... Done
Building dependency tree
Reading state information... Done
packagekit is already the newest version (1.1.13-2ubuntu1.1).
packagekit set to manually installed.
samba-libs is already the newest version (2:4.13.17~dfsg-0ubuntu1.20.04.5).
samba-libs set to manually installed.
The following package was automatically installed and is no longer required:
  gir1.2-goa-1.0
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  krb5-config ldap-utils libbasicobjects0 libc-ares2 libcollection4 libdhash1 libgssrpc4 libini-config5 libipa-hbac0 libkadm5clnt-mit11 libkadm5srv-mit11
  libkdb5-9 libnfsidmap2 libnss-sss libpam-pwquality libpam-sss libpath-utils1 libref-array1 libsassl2-modules-gssapi-mit libsss-certmap0 libsss-idmap0
  libsss-nss-idmap0 libsss-sudo python3-crypto python3-gpg python3-samba python3-sss python3-tdb samba-common samba-dsdb-modules sssd-ad sssd-ad-common
  sssd-common sssd-ipa sssd-krb5 sssd-krb5-common sssd-ldap sssd-proxy
Suggested packages:
  krb5-k5tls krb5-doc heimdal-clients python3-markdown python3-dnspython libsassl2-modules-ldap
The following NEW packages will be installed:
  adcli krb5-config krb5-user ldap-utils libbasicobjects0 libc-ares2 libcollection4 libdhash1 libgssrpc4 libini-config5 libipa-hbac0 libkadm5clnt-mit11
  libkadm5srv-mit11 libkdb5-9 libnfsidmap2 libnss-sss libpam-pwquality libpam-sss libpath-utils1 libref-array1 libsassl2-modules-gssapi-mit libsss-certmap0
  libsss-idmap0 libsss-nss-idmap0 libsss-sudo python3-crypto python3-gpg python3-samba python3-sss python3-tdb realmd samba-common samba-common-bin
  samba-dsdb-modules sssd sssd-ad sssd-ad-common sssd-common sssd-ipa sssd-krb5 sssd-krb5-common sssd-ldap sssd-proxy sssd-tools
0 upgraded, 44 newly installed, 0 to remove and 7 not upgraded.
Need to get 6.701 kB of archives.
After this operation, 39,9 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

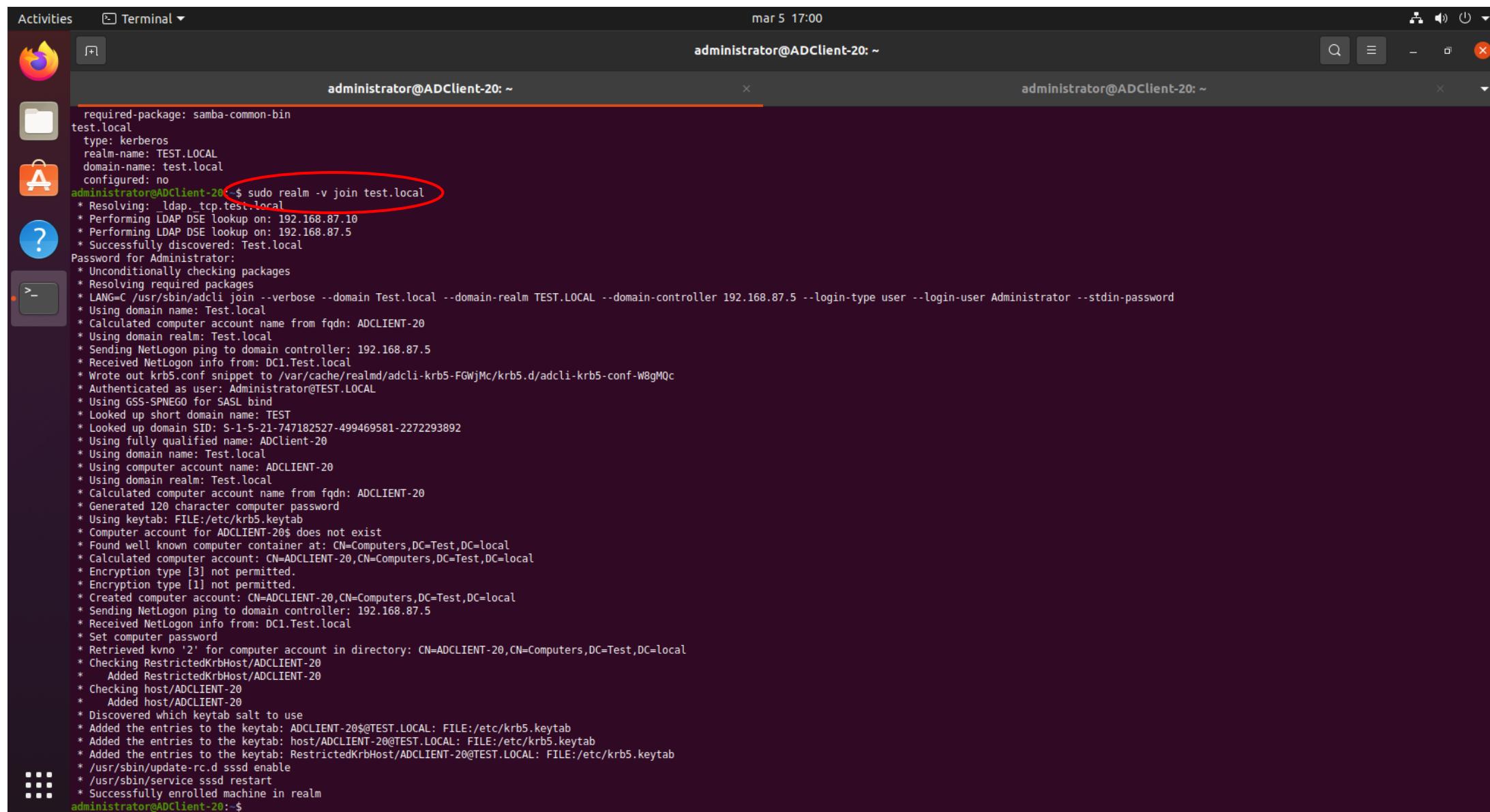
sudo apt install sssd sssd-tools realmd adcli packagekit krb5-user samba-libs samba-common-bin

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```
Activities Terminal ▾ mar 5 16:57
administrator@ADClient-20: ~
administrator@ADClient-20: ~
administrator@ADClient-20:~$ sudo realm -v discover test.local
* Resolving: _ldap._tcp.test.local
* Performing LDAP DSE lookup on: 192.168.87.10
* Performing LDAP DSE lookup on: 192.168.87.5
* Successfully discovered: Test.local
Test.local
type: kerberos
realm-name: TEST.LOCAL
domain-name: Test.local
configured: no
server-software: active-directory
client-software: sssd
required-package: sssd-tools
required-package: sssd
required-package: libnss-sss
required-package: libpam-sss
required-package: adcli
required-package: samba-common-bin
test.local
type: kerberos
realm-name: TEST.LOCAL
domain-name: test.local
configured: no
administrator@ADClient-20:~$
```

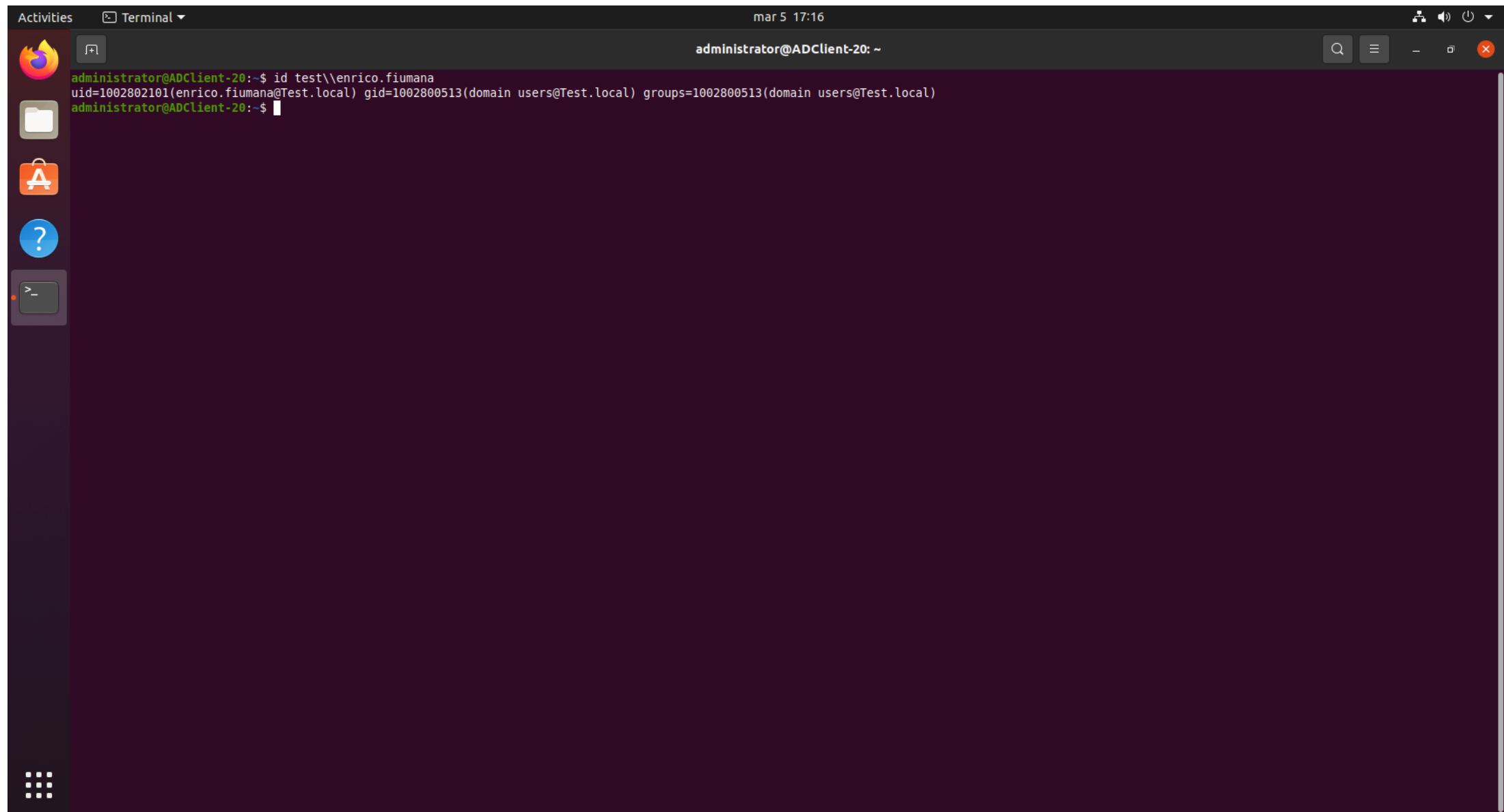

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The screenshot shows a Linux terminal window with a dark theme. The terminal title bar indicates the user is 'administrator@ADClient-20' and the time is 'mar 5 17:00'. The terminal content shows the output of the 'realm join' command, which is used to join a Linux machine to an Active Directory domain. The output includes details about the domain (TEST.LOCAL), the realm (TEST.LOCAL), and the computer account (ADCLIENT-20). The command 'sudo realm -v join test.local' is highlighted with a red circle. The terminal also shows the password for the Administrator user and the successful enrollment of the machine into the realm.

```
required-package: samba-common-bin
test.local
type: kerberos
realm-name: TEST.LOCAL
domain-name: test.local
configured: no
administrator@ADClient-20:~$ sudo realm -v join test.local
* Resolving: ldap.tcp.test.local
* Performing LDAP DSE lookup on: 192.168.87.10
* Performing LDAP DSE lookup on: 192.168.87.5
* Successfully discovered: Test.local
Password for Administrator:
* Unconditionally checking packages
* Resolving required packages
* LANG=C /usr/sbin/adcli join --verbose --domain Test.local --domain-realm TEST.LOCAL --domain-controller 192.168.87.5 --login-type user --login-user Administrator --stdin-password
* Using domain name: Test.local
* Calculated computer account name from fqdn: ADCLIENT-20
* Using domain realm: Test.local
* Sending NetLogon ping to domain controller: 192.168.87.5
* Received NetLogon info from: DC1.Test.local
* Wrote out krb5.conf snippet to /var/cache/realmd/adcli-krb5-FGWjMc/krb5.d/adcli-krb5-conf-W8gMQC
* Authenticated as user: Administrator@TEST.LOCAL
* Using GSS-SPNEGO for SASL bind
* Looked up short domain name: TEST
* Looked up domain SID: S-1-5-21-747182527-499469581-2272293892
* Using fully qualified name: ADClient-20
* Using domain name: Test.local
* Using computer account name: ADCLIENT-20
* Using domain realm: Test.local
* Calculated computer account name from fqdn: ADCLIENT-20
* Generated 120 character computer password
* Using keytab: FILE:/etc/krb5.keytab
* Computer account for ADCLIENT-20$ does not exist
* Found well known computer container at: CN=Computers,DC=Test,DC=local
* Calculated computer account: CN=ADCLIENT-20,CN=Computers,DC=Test,DC=local
* Encryption type [3] not permitted.
* Encryption type [1] not permitted.
* Created computer account: CN=ADCLIENT-20,CN=Computers,DC=Test,DC=local
* Sending NetLogon ping to domain controller: 192.168.87.5
* Received NetLogon info from: DC1.Test.local
* Set computer password
* Retrieved kvno '2' for computer account in directory: CN=ADCLIENT-20,CN=Computers,DC=Test,DC=local
* Checking RestrictedKrbHost/ADCLIENT-20
*   Added RestrictedKrbHost/ADCLIENT-20
* Checking host/ADCLIENT-20
*   Added host/ADCLIENT-20
* Discovered which keytab salt to use
* Added the entries to the keytab: ADCLIENT-20$@TEST.LOCAL: FILE:/etc/krb5.keytab
* Added the entries to the keytab: host/ADCLIENT-20@TEST.LOCAL: FILE:/etc/krb5.keytab
* Added the entries to the keytab: RestrictedKrbHost/ADCLIENT-20@TEST.LOCAL: FILE:/etc/krb5.keytab
* /usr/sbin/update-rc.d sssd enable
* /usr/sbin/service sssd restart
* Successfully enrolled machine in realm
administrator@ADClient-20:~$
```

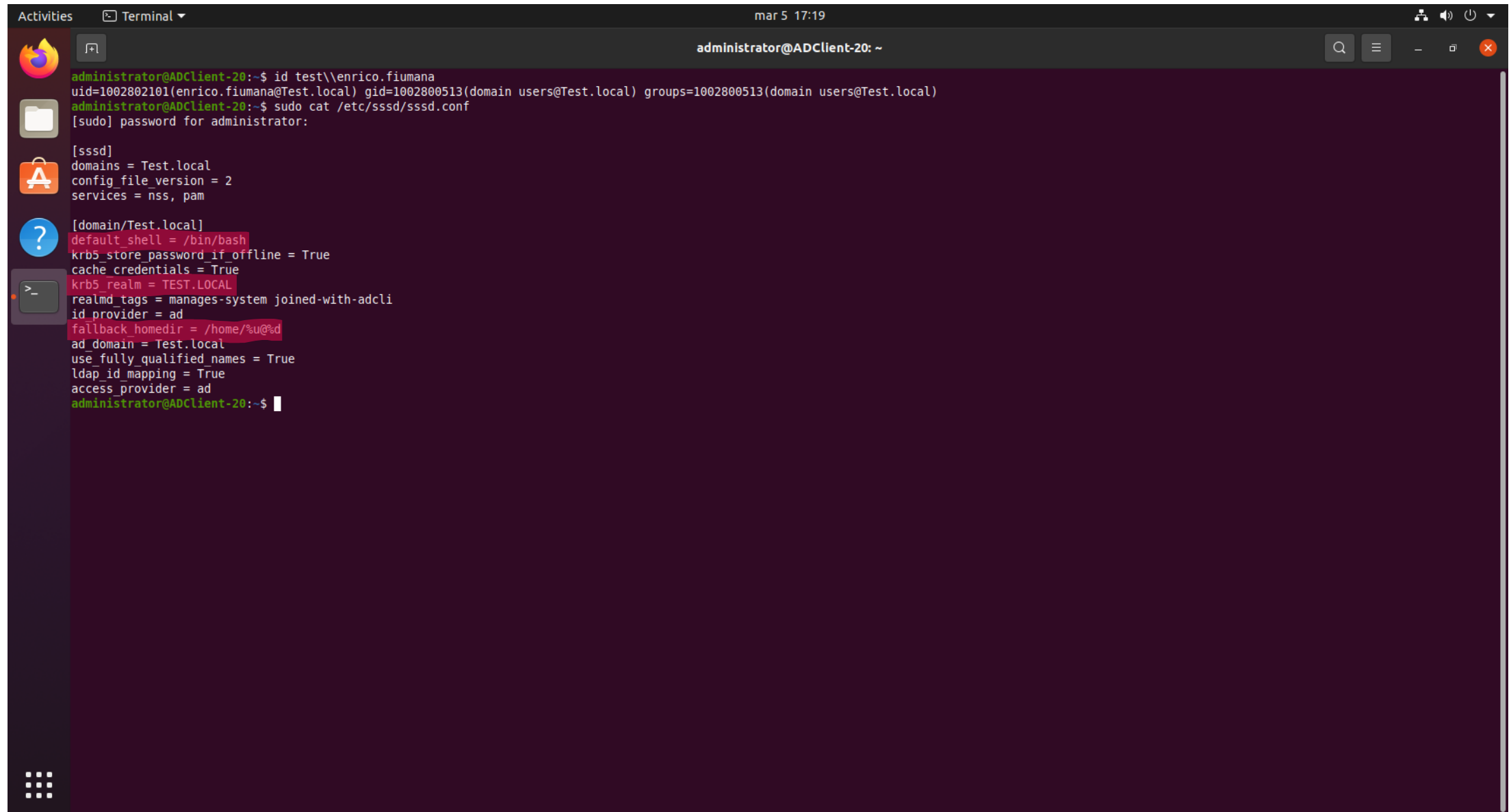
Active Directory e Linux



The screenshot shows a Linux desktop environment with a terminal window open. The terminal title bar reads "administrator@ADClient-20: ~". The terminal content shows the command `id test\\enrico.fiumana` being executed, resulting in the output: `uid=1002802101(enrico.fiumana@Test.local) gid=1002800513(domain users@Test.local) groups=1002800513(domain users@Test.local)`. The terminal window has a dark purple background and a light gray border. The desktop background is dark gray. On the left side, there is a vertical dock with icons for Firefox, a file manager, an application store, a help icon, and a terminal icon. The top of the window shows a status bar with "Activities", "Terminal", and the date/time "mar 5 17:16".

```
administrator@ADClient-20:~$ id test\\enrico.fiumana
uid=1002802101(enrico.fiumana@Test.local) gid=1002800513(domain users@Test.local) groups=1002800513(domain users@Test.local)
administrator@ADClient-20:~$
```


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The screenshot shows a Linux terminal window titled "administrator@ADClient-20: ~". The terminal displays the following commands and output:

```
administrator@ADClient-20:~$ id test\\enrico.fiumana
uid=1002802101(enrico.fiumana@Test.local) gid=1002800513(domain users@Test.local) groups=1002800513(domain users@Test.local)
administrator@ADClient-20:~$ sudo cat /etc/sss/sssd.conf
[sudo] password for administrator:

[sssd]
domains = Test.local
config_file_version = 2
services = nss, pam

[domain/Test.local]
default_shell = /bin/bash
krb5_store_password_if_offline = True
cache_credentials = True
krb5_realm = TEST.LOCAL
realmd_tags = manages-system joined-with-adcli
id_provider = ad
fallback_homedir = /home/%u@%d
ad_domain = Test.local
use_fully_qualified_names = True
ldap_id_mapping = True
access_provider = ad
administrator@ADClient-20:~$
```

The terminal window includes a sidebar with icons for Activities, Terminal, and a file manager. The top bar shows the date and time as "mar 5 17:19".

Active Directory e Linux

Problema:
La home directory non esiste
Come facciamo?

Soluzione:
PAM: Pluggable
Authentication Modules

```
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-60-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

1 device has a firmware upgrade available.
Run 'fwupdmgmt get-upgrades' for more information.

* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.

  https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

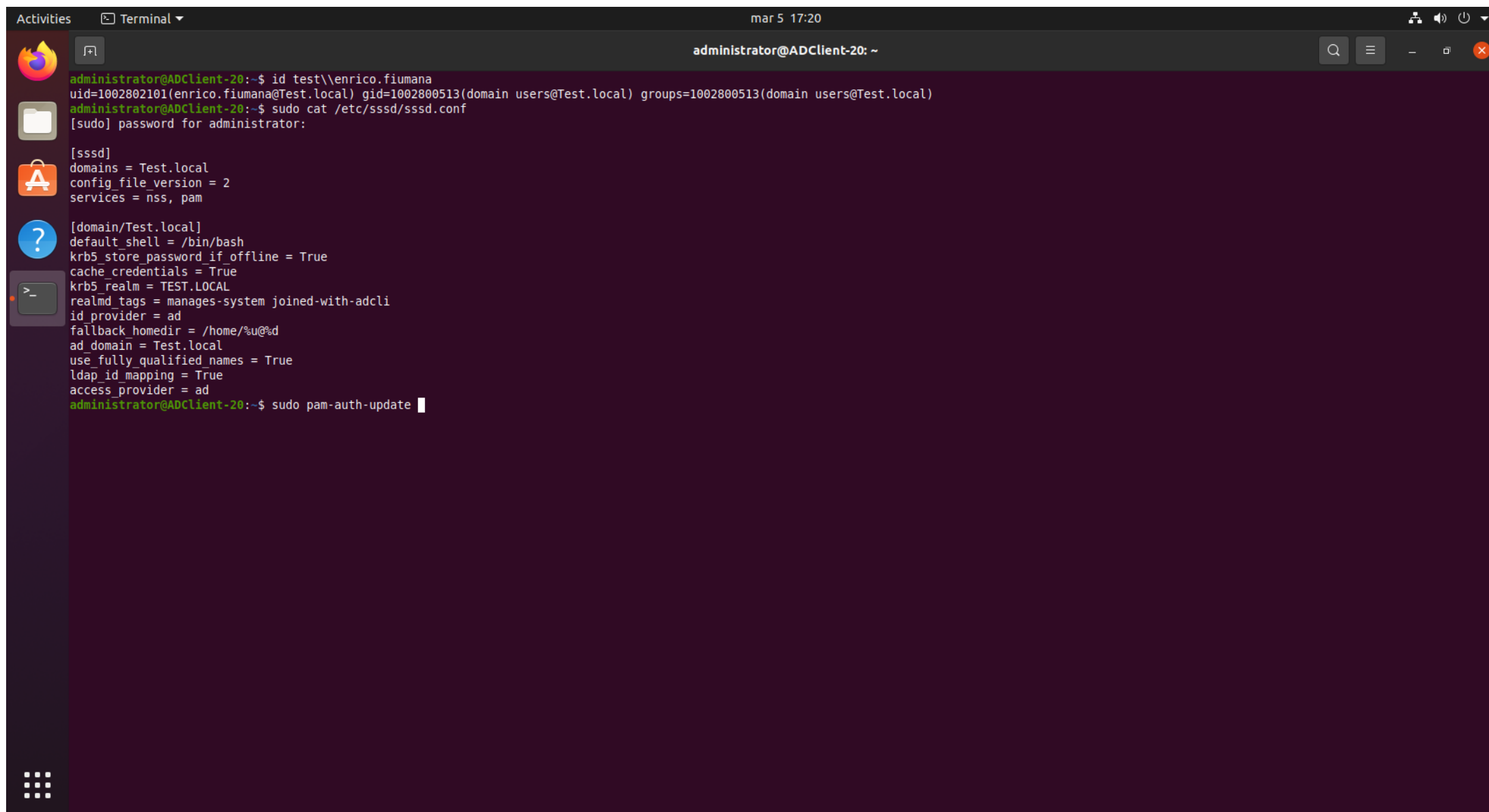
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

No directory, logging in with HOME=/
enrico.fiumana@Test.local@ADClient-20:/$ pwd
/
enrico.fiumana@Test.local@ADClient-20:/$ echo $HOME
/
enrico.fiumana@Test.local@ADClient-20:/$
```

Active Directory e Linux



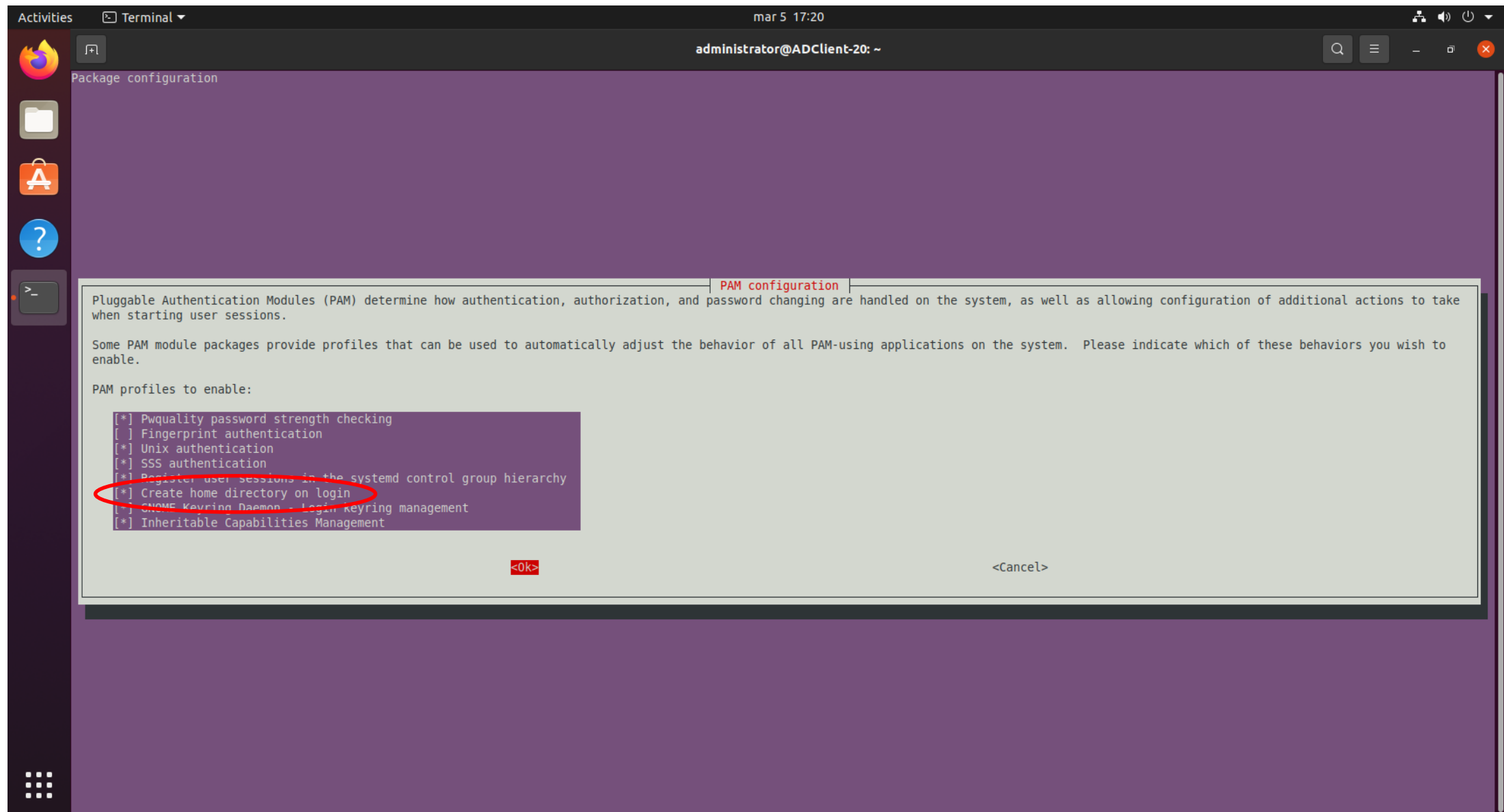
The screenshot shows a Linux terminal window with a dark theme. The window title is "administrator@ADClient-20: ~". The terminal displays the following commands and output:

```
administrator@ADClient-20:~$ id test\enrico.fiumana
uid=1002802101(enrico.fiumana@Test.local) gid=1002800513(domain users@Test.local) groups=1002800513(domain users@Test.local)
administrator@ADClient-20:~$ sudo cat /etc/sss/sssd.conf
[sudo] password for administrator:
[sssd]
domains = Test.local
config_file_version = 2
services = nss, pam

[domain/Test.local]
default_shell = /bin/bash
krb5_store_password_if_offline = True
cache_credentials = True
krb5_realm = TEST.LOCAL
realmd_tags = manages-system joined-with-adcli
id_provider = ad
fallback_homedir = /home/%u@%d
ad_domain = Test.local
use_fully_qualified_names = True
ldap_id_mapping = True
access_provider = ad
administrator@ADClient-20:~$ sudo pam-auth-update
```

The terminal window includes a sidebar on the left with icons for Activities, Terminal, and other applications. The top status bar shows the date and time as "mar 5 17:20".

Active Directory e Linux



Active Directory e Linux

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

1 device has a firmware upgrade available.
Run 'fwupdmgm get-upgrades' for more information.

* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.

  https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***

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individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Last login: Sun Mar  5 17:18:28 CET 2023 on tty3
Creating directory '/home/enrico.fiumana@Test.local'.
enrico.fiumana@Test.local@ADClient-20:~$ pwd
/home/enrico.fiumana@Test.local
enrico.fiumana@Test.local@ADClient-20:~$
```

Active Directory e Linux

Mediante klist visualizzo
alcune in formazione del ticket
kerberos

```
https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Last login: Sun Mar  5 17:18:28 CET 2023 on tty3
Creating directory '/home/enrico.fiumana@Test.local'.
enrico.fiumana@Test.local@ADClient-20:~$ pwd
/home/enrico.fiumana@Test.local
enrico.fiumana@Test.local@ADClient-20:~$
enrico.fiumana@Test.local@ADClient-20:~$
enrico.fiumana@Test.local@ADClient-20:~$
enrico.fiumana@Test.local@ADClient-20:~$
enrico.fiumana@Test.local@ADClient-20:~$
enrico.fiumana@Test.local@ADClient-20:~$
enrico.fiumana@Test.local@ADClient-20:~$ klist
Ticket cache: FILE:/tmp/krb5cc_1002802101_h4kY0H
Default principal: enrico.fiumana@TEST.LOCAL

Valid starting    Expires          Service principal
05/03/2023 17:20:53 06/03/2023 03:20:53  krbtgt/TEST.LOCAL@TEST.LOCAL
    renew until 06/03/2023 17:20:53
enrico.fiumana@Test.local@ADClient-20:~$
```

Active Directory e Linux

Gli utenti AD sono per Linux utenti «normali», cioè non possono eseguire il comando «**sudo**»

Mediante sudo posso eseguire comandi come se fossi l'utente root

Questo serve agli amministratori di sistema per poter utilizzare normalmente il proprio account e solo quando necessario elevarsi a root, evitando così problemi dovuti alla disattenzione

Il comando sudo utilizza un file di configurazione chiamato «**sudoers**» che si trova in /etc. All'interno vengono riportati gli account e/o i gruppi degli utenti che possono eseguire sudo e quali eventuali comandi possono essere eseguiti.

Ora creiamo un gruppo su AD dentro al quale metteremo gli utenti che vogliamo possano fare sudo poi andremo ad aggiungere tale gruppo al file /etc/sudoers

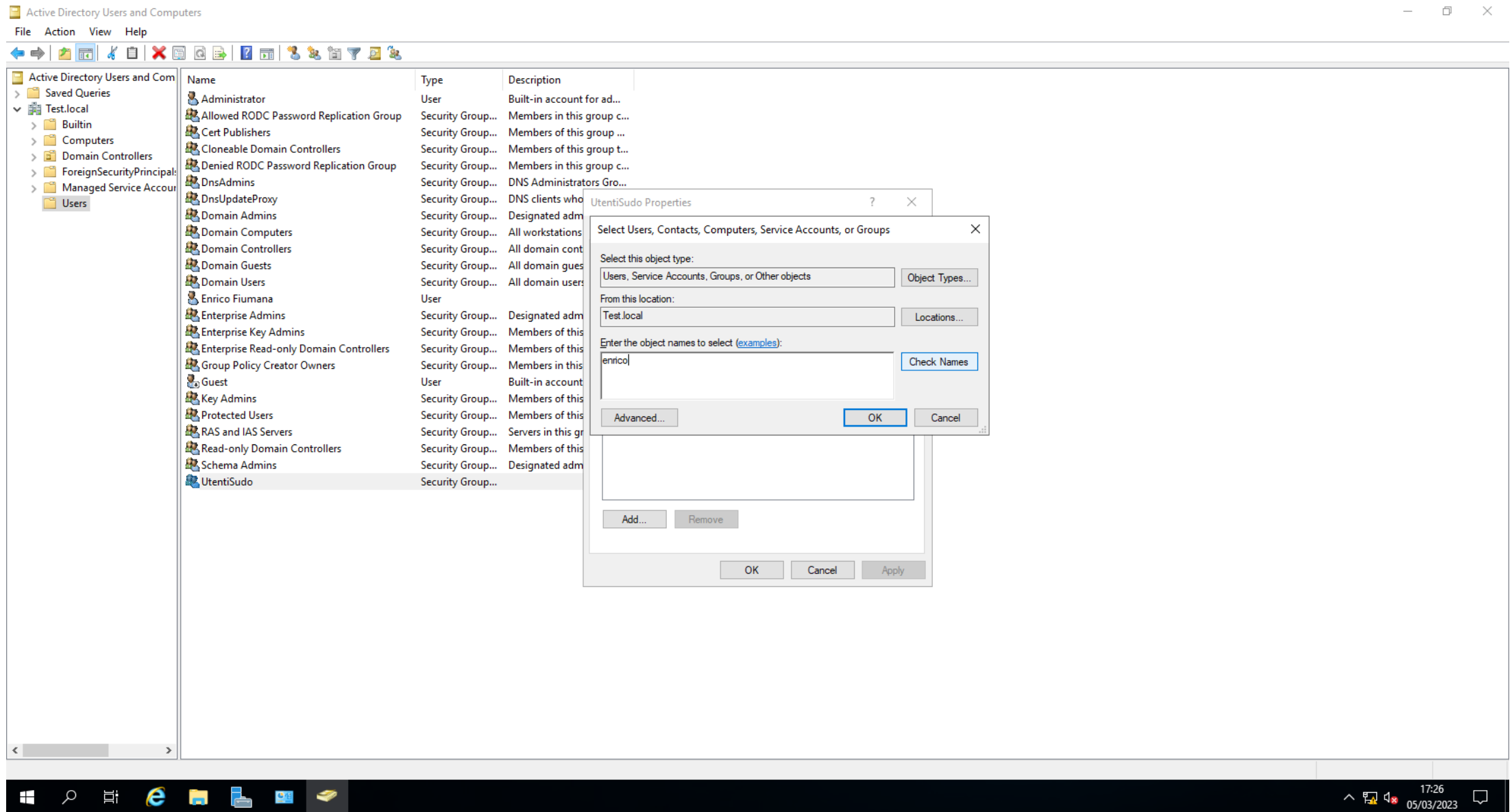
Active Directory e Linux

The screenshot displays the 'Active Directory Users and Computers' console window. The left pane shows the tree structure with 'Test.local' expanded, and 'Users' selected. The main pane lists various groups, with 'UtentiSudo' highlighted. A 'UtentiSudo Properties' dialog box is open, showing the 'General' tab. The dialog contains the following fields and options:

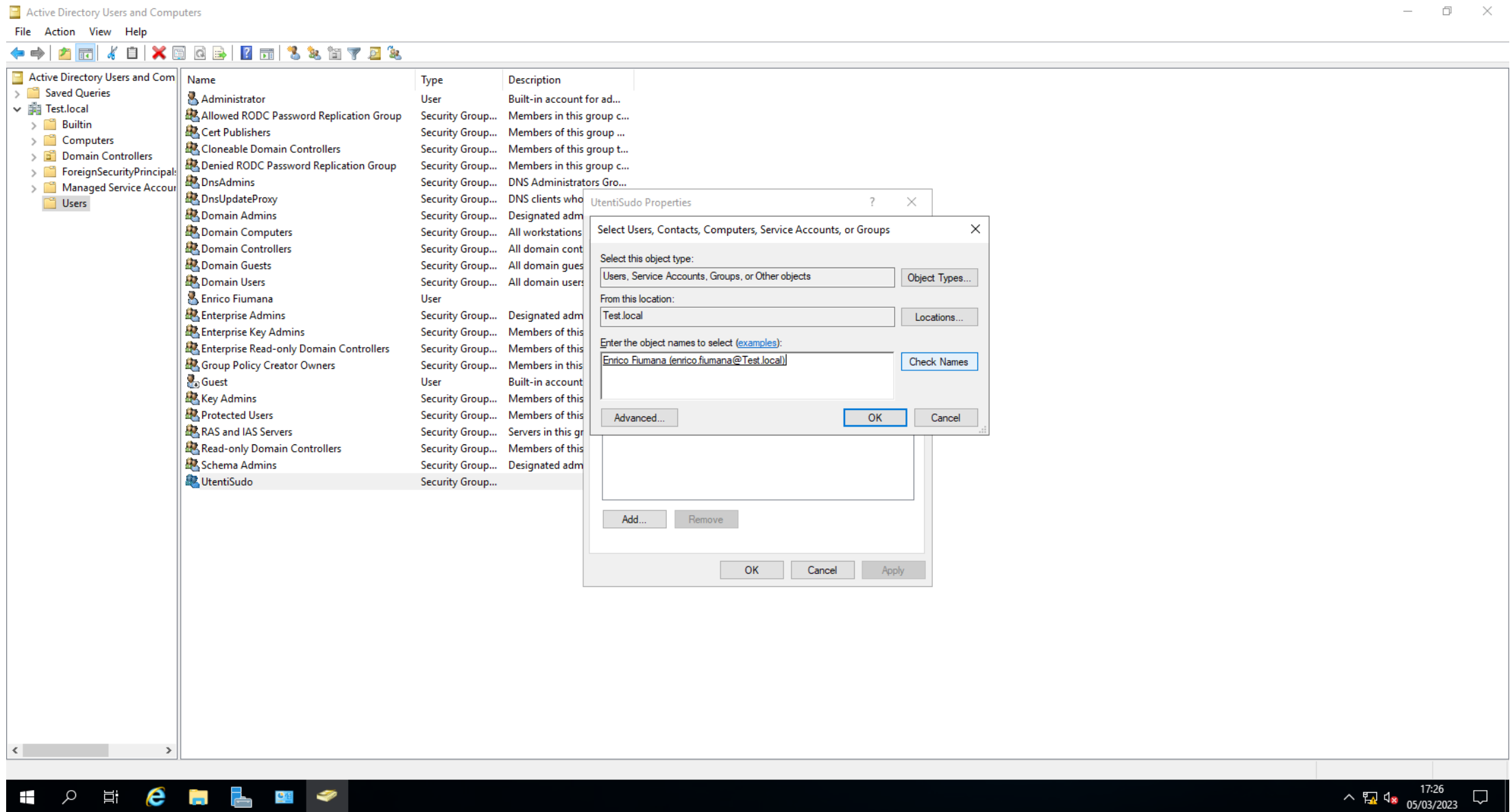
- Group name (pre-Windows 2000):** UtentiSudo
- Description:** (empty text box)
- E-mail:** (empty text box)
- Group scope:** ☒ Domain local, ☐ Global, ☐ Universal
- Group type:** ☒ Security, ☐ Distribution
- Notes:** (empty text box)

The Windows taskbar at the bottom shows the time as 17:25 on 05/03/2023.

Active Directory e Linux



Active Directory e Linux



Active Directory e Linux

The screenshot displays the 'Active Directory Users and Computers' console window. The left pane shows the tree structure with 'Test.local' expanded, and 'Users' selected. The main pane lists domain users and groups. The 'UtentiSudo' group is selected, and its properties dialog is open, showing the 'Members' tab. The 'Members' list contains one entry: 'Enrico Fiumana' from the 'Test.local/Users' folder. The 'Add...' button is visible below the list.

Name	Type	Description
Administrator	User	Built-in account for ad...
Allowed RODC Password Replication Group	Security Group...	Members in this group c...
Cert Publishers	Security Group...	Members of this group ...
Cloneable Domain Controllers	Security Group...	Members of this group t...
Denied RODC Password Replication Group	Security Group...	Members in this group c...
DnsAdmins	Security Group...	DNS Administrators Gro...
DnsUpdateProxy	Security Group...	DNS clients who...
Domain Admins	Security Group...	Designated adm...
Domain Computers	Security Group...	All workstations
Domain Controllers	Security Group...	All domain cont...
Domain Guests	Security Group...	All domain gues...
Domain Users	Security Group...	All domain users
Enrico Fiumana	User	
Enterprise Admins	Security Group...	Designated adm...
Enterprise Key Admins	Security Group...	Members of this...
Enterprise Read-only Domain Controllers	Security Group...	Members of this...
Group Policy Creator Owners	Security Group...	Members in this...
Guest	User	Built-in account...
Key Admins	Security Group...	Members of this...
Protected Users	Security Group...	Members of this...
RAS and IAS Servers	Security Group...	Servers in this gr...
Read-only Domain Controllers	Security Group...	Members of this...
Schema Admins	Security Group...	Designated adm...
UtentiSudo	Security Group...	

UtentiSudo Properties

General | Members | Member Of | Managed By

Members:

Name	Active Directory Domain Services Folder
Enrico Fiumana	Test.local/Users

Add... Remove

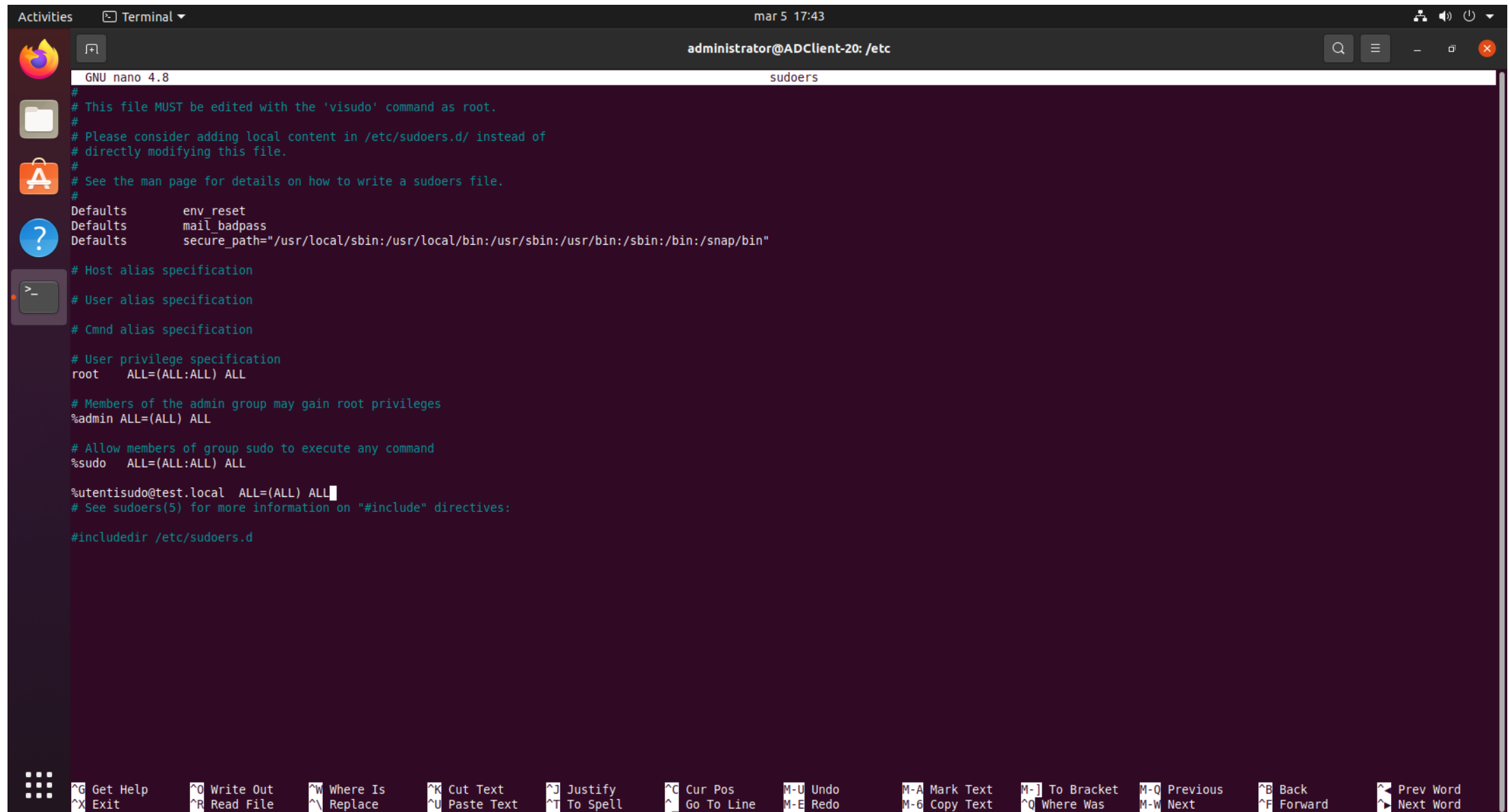
OK Cancel Apply

Active Directory e Linux

The screenshot shows the 'Active Directory Users and Computers' console. The left pane shows the tree structure with 'Test.local' expanded and 'Users' selected. The main pane displays a list of users and groups. The 'NoSudo' user is highlighted. A 'NoSudo Properties' dialog box is open, showing the 'General' tab. The 'Member of' list contains 'Domain Users' and 'Test.local/Users'. The 'Primary group' is set to 'Domain Users'. The 'Set Primary Group' button is disabled with a message: 'There is no need to change Primary group unless you have Macintosh clients or POSIX-compliant applications.'

Name	Type	Description
Administrator	User	Built-in account for ad...
Allowed RODC Password Replication Group	Security Group...	Members in this group c...
Cert Publishers	Security Group...	Members of this group ...
Cloneable Domain Controllers	Security Group...	Members of this group t...
Denied RODC Password Replication Group	Security Group...	Members in this group c...
DnsAdmins	Security Group...	DNS Administrators Gro...
DnsUpdateProxy	Security Group...	DNS clients who...
Domain Admins	Security Group...	Designated adm...
Domain Computers	Security Group...	All workstations
Domain Controllers	Security Group...	All domain cont...
Domain Guests	Security Group...	All domain gues...
Domain Users	Security Group...	All domain users
Enrico Fiumana	User	
Enterprise Admins	Security Group...	Designated adm...
Enterprise Key Admins	Security Group...	Members of this
Enterprise Read-only Domain Controllers	Security Group...	Members of this
Group Policy Creator Owners	Security Group...	Members in this
Guest	User	Built-in account
Key Admins	Security Group...	Members of this
NoSudo	User	
Protected Users	Security Group...	Members of this
RAS and IAS Servers	Security Group...	Servers in this gr
Read-only Domain Controllers	Security Group...	Members of this
Schema Admins	Security Group...	Designated adm...
UtentiSudo	Security Group...	

Active Directory e Linux



The screenshot shows a Linux terminal window with the title bar "Activities Terminal" and the system clock "mar 5 17:43". The terminal is running the nano text editor, editing the file `/etc/sudoers` as the `administrator@ADClient-20` user. The editor's status bar at the top indicates "GNU nano 4.8" and "sudoers".

The content of the `/etc/sudoers` file is as follows:

```
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

%utentisudo@test.local  ALL=(ALL) ALL
# See sudoers(5) for more information on "#include" directives:

#include_dir /etc/sudoers.d
```

The terminal window includes a sidebar on the left with icons for Activities, Terminal, and a file manager. At the bottom, a detailed keyboard shortcuts menu is visible, listing commands such as `^G` Get Help, `^O` Write Out, `^W` Where Is, `^K` Cut Text, `^J` Justify, `^C` Cur Pos, `M-U` Undo, `M-A` Mark Text, `M-I` To Bracket, `M-Q` Previous, `^B` Back, `^F` Forward, `^X` Exit, `^R` Read File, `^N` Replace, `^U` Paste Text, `^T` To Spell, `^L` Go To Line, `M-E` Redo, `M-6` Copy Text, `^Q` Where Was, `M-W` Next, and `^P` Prev Word, `^N` Next Word.

Active Directory e Linux

```
Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***
Last login: Sun Mar  5 17:37:59 CET 2023 on tty3
enrico.fiumana@Test.local@ADClient-20:~$ sudo cat /etc/sudoers
[sudo] password for enrico.fiumana@Test.local:
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

%utentisudo@test.local  ALL=(ALL) ALL
# See sudoers(5) for more information on "#include" directives:

#include_dir /etc/sudoers.d
enrico.fiumana@Test.local@ADClient-20:~$
```

Active Directory e Linux

```
Password:
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-60-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

   https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Creating directory '/home/nosudo@Test.local'.
nosudo@Test.local@ADClient-20:~$
```

Active Directory e Linux

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

1 device has a firmware upgrade available.
Run 'fwupdmgmt get-upgrades' for more information.

* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
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  https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***

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The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Creating directory '/home/nosudo@Test.local'.
nosudo@Test.local@ADClient-20:~$ sudo cat /etc/sudoers
[sudo] password for nosudo@Test.local:
nosudo@Test.local is not allowed to run sudo on ADClient-20. This incident will be reported.
nosudo@Test.local@ADClient-20:~$
```


Active Directory e Linux

Ora andremo ad utilizzare il ticket kerberos per autenticarci sul servizio di condivisione file del domain controller.

Per far questo utilizzeremo il programma «smbclient» che permette di visualizzare/utilizzare le condivisioni SMB (Server Message Block), cioè le condivisioni di file di Windows

```
Ubuntu 20.04.5 LTS ADClient-20 tty3
ADClient-20 login:
ADClient-20 login: test\enrico.fiumana
Password:
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-60-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

1 device has a firmware upgrade available.
Run 'fwupdmgmt get-upgrades' for more information.

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7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***
Last login: Sun Mar  5 17:41:47 CET 2023 on tty3
enrico.fiumana@Test.local@ADClient-20:~$ smbclient

Command 'smbclient' not found, but can be installed with:

apt install smbclient
Please ask your administrator.

enrico.fiumana@Test.local@ADClient-20:~$ sudo apt install smbclient_
```

Active Directory e Linux

```
Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***
Last login: Sun Mar  5 17:41:47 CET 2023 on tty3
enrico.fiumana@Test.local@ADClient-20:~$ smbclient

Command 'smbclient' not found, but can be installed with:

apt install smbclient
Please ask your administrator.

enrico.fiumana@Test.local@ADClient-20:~$ sudo apt install smbclient
[sudo] password for enrico.fiumana@Test.local:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
  gir1.2-goa-1.0
Use 'sudo apt autoremove' to remove it.
Suggested packages:
  cifs-utils heimdal-clients
The following NEW packages will be installed:
  smbclient
0 upgraded, 1 newly installed, 0 to remove and 7 not upgraded.
Need to get 402 kB of archives.
After this operation, 2.103 kB of additional disk space will be used.
Get:1 http://it.archive.ubuntu.com/ubuntu focal-updates/main amd64 smbclient amd64 2:4.13.17~dfsg-0ubuntu1.20.04.5 [402 kB]
Fetched 402 kB in 0s (1.321 kB/s)
Selecting previously unselected package smbclient.
(Reading database ... 158812 files and directories currently installed.)
Preparing to unpack .../smbclient_2%3a4.13.17~dfsg-0ubuntu1.20.04.5_amd64.deb ...
Unpacking smbclient (2:4.13.17~dfsg-0ubuntu1.20.04.5) ...
Setting up smbclient (2:4.13.17~dfsg-0ubuntu1.20.04.5) ...
Processing triggers for man-db (2.9.1-1) ...
enrico.fiumana@Test.local@ADClient-20:~$ _
```

Active Directory e Linux

```
New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
*** System restart required ***
Last login: Sun Mar  5 17:41:47 CET 2023 on tty3
enrico.fiumana@Test.local@ADClient-20:~$ smbclient

Command 'smbclient' not found, but can be installed with:

apt install smbclient
Please ask your administrator.

enrico.fiumana@Test.local@ADClient-20:~$ sudo apt install smbclient
[sudo] password for enrico.fiumana@Test.local:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
  gir1.2-goa-1.0
Use 'sudo apt autoremove' to remove it.
Suggested packages:
  cifs-utils heimdal-clients
The following NEW packages will be installed:
  smbclient
0 upgraded, 1 newly installed, 0 to remove and 7 not upgraded.
Need to get 402 kB of archives.
After this operation, 2.103 kB of additional disk space will be used.
Get:1 http://it.archive.ubuntu.com/ubuntu focal-updates/main amd64 smbclient amd64 2:4.13.17~dfsg-0ubuntu1.20.04.5 [402 kB]
Fetched 402 kB in 0s (1.321 kB/s)
Selecting previously unselected package smbclient.
(Reading database ... 158812 files and directories currently installed.)
Preparing to unpack .../smbclient_2%3a4.13.17~dfsg-0ubuntu1.20.04.5_amd64.deb ...
Unpacking smbclient (2:4.13.17~dfsg-0ubuntu1.20.04.5) ...
Setting up smbclient (2:4.13.17~dfsg-0ubuntu1.20.04.5) ...
Processing triggers for man-db (2.9.1-1) ...
enrico.fiumana@Test.local@ADClient-20:~$ smbclient -k -L dc1.test.local

      Sharename      Type      Comment
      -----
      ADMIN$         Disk      Remote Admin
      C$              Disk      Default share
      IPC$            IPC       Remote IPC
      NETLOGON        Disk      Logon server share
      SYSVOL          Disk      Logon server share
SMB1 disabled -- no workgroup available
enrico.fiumana@Test.local@ADClient-20:~$
```